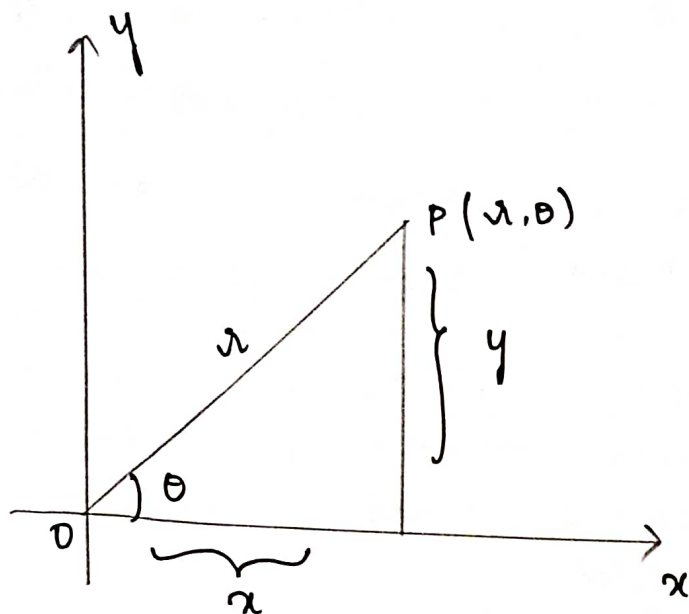


12 | 8 | 2022

Unit - 1

Differential Calculus

r = radial distance
 OP = radius vector
 θ = polar angle

Consider a point 'P' in the x - y plane. Join the pt 'O' and 'P'. Let ' r ' be the length of the line segment OP. And ' θ ' be the angle which OP makes with the x -axis. Then (r, θ) is called the polar co-ordinates of pt 'P'.

We write $P = (r, \theta)$

or $P(r, \theta)$

In particular ' r ' is called radial distance, ' θ ' - polar angle and 'O' is called the pole. And the x -axis is called initial line.

The line segment OP is called radius vector.

Let (x, y) be the cartesian co-ordination of the pt 'P'.

We have $x = r \cos \theta$

$y = r \sin \theta$ then.

$$r = \sqrt{x^2 + y^2}$$

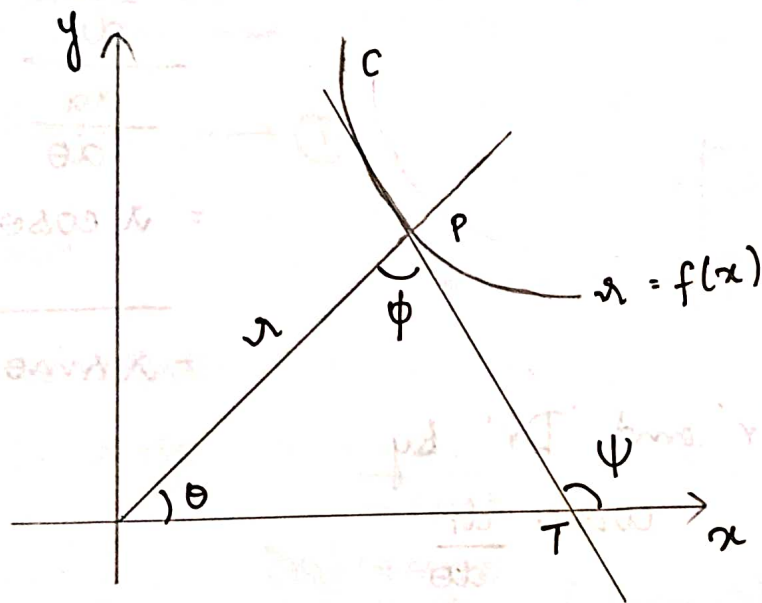
$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$

w.k.t the eqⁿ of the plane curve 'C' in the cartesian form is $y = f(x)$ called cartesian curve. Similarly the eqⁿ 'C' in the polar form.

$r = f(\theta)$ is called polar curve.

* Angle between radius vector and tangent: *

If ϕ be the angle b/w radius and vector and the tangent at any point on the curve $r = f(\theta)$ then $\tan \phi = r \frac{d\theta}{dr}$.



Proof: Let 'P' be the point on the curve $r = f(\theta)$

Draw a tangent line passes through the point 'P' touches the x-axis at T. Join OP and ϕ be the angle b/w radius vector and tangent line PT. Let ψ be the angle b/w x-axis and the tangent line.

We know that $\psi = \theta + \phi$

$$\begin{aligned} \tan \psi &= \tan (\theta + \phi) \quad (\because \tan(A+B) \text{ formula}) \\ &= \frac{\tan \theta + \tan \phi}{1 - \tan \theta \tan \phi} \quad \text{--- (1)} \end{aligned}$$

We have $x = r \cos \theta$, $y = r \sin \theta$

$$\frac{dx}{d\theta} = -r \sin \theta + \cos \theta \frac{dr}{d\theta}$$

$$\frac{dy}{d\theta} = r \cos \theta + \sin \theta \frac{dr}{d\theta}$$

We know that, $\tan \phi = \frac{dy}{dx}$

$$= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$= \frac{r \cos \theta + \sin \theta \frac{dr}{d\theta}}{-r \sin \theta + \cos \theta \frac{dr}{d\theta}}$$

Divide 'Nr' and 'Dr' by

$$\cos \theta \cdot \frac{dr}{d\theta}$$

$$\therefore \tan \phi = \frac{r \cos \theta}{\cos \theta \cdot \frac{dr}{d\theta}} + \frac{\sin \theta \frac{dr}{d\theta}}{\cos \theta \cdot \frac{dr}{d\theta}}$$

$$= \frac{-r \sin \theta}{\cos \theta \cdot \frac{dr}{d\theta}} + \frac{\cos \theta \frac{dr}{d\theta}}{\cos \theta \frac{dr}{d\theta}}$$

$$= \frac{r \frac{d\theta}{dr} + \tan \theta}{-r \tan \theta \frac{d\theta}{dr} + 1}$$

$$= \frac{\tan \theta + r \frac{d\theta}{dr}}{1 - \tan \theta \cdot r \frac{d\theta}{dr}} \quad \text{--- (2)}$$

Comparing (1) and (2) we get .

$$\therefore \boxed{\tan \phi = r \frac{d\theta}{dr}}$$

> For the following curve find angle b/w radius vector and tangent:

1) $r = a(1 + \cos \theta) \rightarrow \text{--- (1)}$

$$\frac{dr}{d\theta} = -a \sin \theta$$

$$\therefore \boxed{\tan \phi = r \frac{d\theta}{dr}}$$

We know that

$$\tan \phi = r \frac{d\theta}{dr} = \frac{r}{-a \sin \theta}$$

$$\tan \phi = \frac{a(1 + \cos \theta)}{-a \sin \theta} \quad (\text{from (1)})$$

$$\tan \phi = \frac{-(1 + \cos \theta)}{\sin \theta}$$

$$= \frac{-2 \cos^2 \theta/2}{2 \sin \theta/2 \cos \theta/2} = -\cot \theta/2$$

$$\tan \phi = -\cot \theta/2$$

$$\tan \phi = \tan \left(\pi/2 + \theta/2 \right)$$

$$\therefore \boxed{\phi = \pi/2 + \theta/2}$$

$$2) \quad r^2 \cos 2\theta = a^2$$

$$2r \frac{dr}{d\theta} \cos 2\theta + r^2 (-\sin 2\theta) \cdot 2 = 0.$$

$$2r \frac{dr}{d\theta} \cos 2\theta - r^2 \sin 2\theta \cdot 2 = 0.$$

$$2r \frac{dr}{d\theta} \cos 2\theta = r^2 \sin 2\theta \cdot 2$$

$$\frac{dr}{d\theta} = \frac{r^2 \sin 2\theta \cdot 2}{2r \cos 2\theta} = r \tan 2\theta.$$

$$\text{Now } \tan \phi = \frac{r}{r \tan 2\theta}$$

$$\tan \phi = \cot 2\theta$$

$$\tan \phi = \tan \left(\frac{\pi}{2} + 2\theta \right)$$

$$\therefore \boxed{\phi = \frac{\pi}{2} + 2\theta}$$

$$3) \quad r = a e^{\theta \cot \alpha}$$

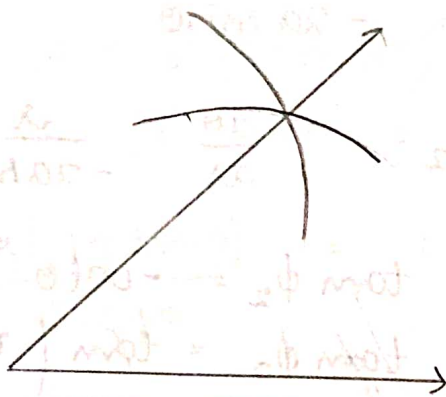
$$\frac{dr}{d\theta} = a e^{\theta \cot \alpha} \cdot \cot \alpha$$

$$= \cot \alpha$$

$$\tan \phi = \frac{r}{\cot \alpha}$$

$$\tan \phi = \frac{a e^{\theta \cot \alpha}}{\cot \alpha}$$

7 Angle b/w polar curves:



Suppose 2 curves ' C_1 ' and ' C_2 ' intersect at a pt ' P ' then angle of intersection b/w C_1 and C_2 at ' P ' is the angle b/w the tangents Pt_1 and Pt_2 to C_1 and C_2 respectively.

Suppose the radius vector ' OP ' makes angles ' ϕ_1 ' and ' ϕ_2 ' with Pt_1 and Pt_2 respectively then the difference $|\phi_1 - \phi_2|$ is the angle b/w Pt_1 and Pt_2 .

If $\tan \phi_1 \cdot \tan \phi_2 = -1$ then we say that the curves intersect orthogonally.

> Find the angle of intersection b/w following curve:

1. $r = a(1 - \cos \theta)$, $r = 2a \cos \theta$.

For the curve $r = a(1 - \cos \theta)$

$$\frac{dr}{d\theta} = a \sin \theta$$

$$\tan \phi_1 = r \frac{d\theta}{dr} = r \frac{1}{a \sin \theta} = \frac{a(1 - \cos \theta)}{a \sin \theta} = \frac{2 \sin^2 \theta/2}{2 \sin \theta/2 \cos \theta/2}$$

$$\tan \phi_1 = \tan \theta/2$$

$$\therefore \boxed{\phi_1 = \theta/2}$$

For the curve $r = 2a \cos \theta$

$$\frac{dr}{d\theta} = -2a \sin \theta$$

$$\tan \phi_2 = r \frac{d\theta}{dr} = \frac{r}{-2a \sin \theta} = \frac{2a \cos \theta}{-2a \sin \theta} =$$

$$\tan \phi_2 = -\cot \theta$$

$$\tan \phi_2 = \tan \left(\frac{\pi}{2} + \theta \right)$$

$$\therefore \boxed{\phi_2 = \frac{\pi}{2} + \theta}$$

$$\therefore |\phi_1 - \phi_2| = \left| \frac{\theta}{2} - \frac{\pi}{2} - \theta \right|$$

$$= \frac{\pi}{2} + \frac{\theta}{2} //$$

2. $r = 2 \sin \theta$, $r = \sin \theta + \cos \theta$.

$$\rightarrow \frac{dr}{d\theta} = 2 \cos \theta$$

$$\tan \phi_1 = \frac{r}{2 \cos \theta} = \frac{2 \sin \theta}{2 \cos \theta} = \tan \theta$$

$$\therefore \boxed{\phi_1 = \theta}$$

$$\rightarrow \frac{dr}{d\theta} = \sin \theta + \cos \theta$$

$$r\theta = \cos \theta - \sin \theta$$

$$\tan \phi_2 = \frac{r}{\cos \theta - \sin \theta} = \frac{\sin \theta + \cos \theta}{\cos \theta - \sin \theta} \div \cos \theta$$

$$= \frac{\tan \theta + 1}{1 - \tan \theta} = \tan \left(\frac{\pi}{4} + \theta \right)$$

$$\therefore \boxed{\phi_2 = \frac{\pi}{4} + \theta}$$

$$|\phi_1 - \phi_2| = \left| \theta - \frac{\pi}{4} + \theta \right| = \pi/4 //$$

$$3. r = a(1 + \sin \theta), \quad u = b(1 - \sin \theta)$$

$$\rightarrow \frac{dr}{d\theta} = a \cos \theta$$

$$\tan \phi = \frac{u}{a \cos \theta} = \frac{a(1 + \sin \theta)}{a \cos \theta} = \frac{1 + \cos(\pi/2 - \theta)}{\sin(\pi/2 - \theta)} = 2 \cos^2 \left(\frac{\pi/2 - \theta}{2} \right)$$

$$\rightarrow \frac{du}{d\theta} = -b \cos \theta$$

$$\tan \phi = \frac{u}{-b \cos \theta} = \frac{b(1 - \sin \theta)}{-b \cos \theta} =$$

$$= \frac{1 - \cos(\pi/2 - \theta)}{-\sin(\pi/2 - \theta)}$$

$$= \frac{2 \sin^2 \left(\frac{\pi/2 - \theta}{2} \right)}{-2 \sin \left(\frac{\pi/2 - \theta}{2} \right) \cos \left(\frac{\pi/2 - \theta}{2} \right)}$$

$$= -\tan \left(\frac{\pi - 2\theta}{4} \right) = -\tan \left(\pi/4 - \theta/2 \right)$$

$$\phi_2 = -\left(\frac{\pi}{4} - \frac{2\theta}{2} \right) //$$

$$\begin{aligned} &= \cot \left(\frac{\pi/2 - \theta}{4} \right) \cdot \frac{2 \sin \left(\frac{\pi/2 - \theta}{2} \right) \cos \left(\frac{\pi/2 - \theta}{2} \right)}{\left(\frac{\pi/2 - \theta}{2} \right)} \\ &= \tan \left(\frac{\pi/2 - \theta}{4} \right) \cdot \frac{2 \sin \left(\frac{\pi/2 - \theta}{2} \right) \cos \left(\frac{\pi/2 - \theta}{2} \right)}{\left(\frac{\pi/2 - \theta}{2} \right)} \\ &\phi = \left(\frac{\pi}{4} + \frac{\theta}{2} \right) \end{aligned}$$

$$4. \quad r = \frac{a\theta}{1+\theta} \quad \text{and} \quad x = \frac{a}{1+\theta^2}$$

For the curve $r = \frac{a\theta}{1+\theta}$

$$\frac{dr}{d\theta} = \frac{(1+\theta) \cdot a - a\theta}{(1+\theta)^2}$$

$$\tan \phi_1 = r \frac{d\theta}{dr} = \frac{a\theta}{(1+\theta)^2} \times \left[\frac{(1+\theta) - \theta}{1+\theta} \right] a \text{ omit}$$

$$= \frac{a\theta}{1+\theta} \times \frac{(1+\theta)^2}{a}$$

$$\tan \phi_1 = \theta(1+\theta) \rightarrow \textcircled{1}$$

* $\phi_1 \rightarrow$

For the curve $r = \frac{a}{1+\theta^2}$

$$\frac{dr}{d\theta} = \frac{(1+\theta^2) \cdot a - 2a\theta}{(1+\theta^2)^2}$$

$$= \frac{-2a\theta}{(1+\theta^2)^2}$$

$$\tan \phi_2 = r \frac{d\theta}{dr} = \frac{a}{1+\theta^2} \times \frac{(1+\theta^2)^2}{2a\theta}$$

$$= \frac{-(1+\theta^2)}{2\theta} \rightarrow \textcircled{2}$$

$$\tan(\phi_1 - \phi_2) = \frac{\tan \phi_1 - \tan \phi_2}{1 + \tan \phi_1 \tan \phi_2} = \frac{\theta(1+\theta) + \frac{(1+\theta^2)}{2\theta}}{1 + \theta(1+\theta) \left(\frac{1+\theta^2}{2\theta} \right)}$$

At the point of intersection,

$$\frac{a\theta}{1+\theta} \neq \frac{a}{1+\theta^2}$$

$$\theta + \theta^3 = 1 + \theta$$

$$\theta^3 = 1$$

$$\therefore \sqrt{\theta = 1}$$

$$(3) \rightarrow \tan(\phi_1 - \phi_2) = \frac{2+1}{1-2} = -3$$

$$(\phi_1 - \phi_2) = \tan^{-1}(3)$$

5. $r = a \log \theta$ $r = \frac{a}{\log \theta}$

For the curve $r = a \log \theta$

$$\frac{dr}{d\theta} = \frac{a}{\theta}$$

$$\tan \phi_1 = r \frac{d\theta}{dr} = \frac{r \times \theta}{a} \rightarrow \textcircled{1} \quad \frac{a \log \theta \times a}{a}$$

For the curve $r = \frac{a}{\log \theta}$

$$\rightarrow \theta \log \theta \rightarrow \textcircled{1}$$

$$\frac{dr}{d\theta} = \frac{\log \theta \cdot -\frac{1}{\theta} a}{(\log \theta)^2} = \frac{-a}{\theta \log^2 \theta}$$

$$\tan \phi_2 = r \frac{d\theta}{dr} = \frac{-r a}{\theta \log^2 \theta} = \frac{-r \times \theta \log \theta}{\log \theta \cdot a}$$

$$= -\theta \log \theta \rightarrow \textcircled{2}$$

b. Show that the following curves intersect each other orthogonally.

a. $r = a(1 + \cos \theta)$, $r = b(1 - \cos \theta)$

→ For the curve $\frac{dr}{d\theta} = a(1 + \cos \theta)$
 $r = -a \sin \theta$

$$\begin{aligned} \tan \phi_1 &= \frac{r \frac{d\theta}{dr}}{-a \sin \theta} = \frac{r}{-a \sin \theta} = \frac{a(1 + \cos \theta)}{-a \sin \theta} \\ &= \frac{-2 \cos^2 \theta/2}{2 \sin \theta/2 \cos \theta/2} \\ &= -\cot \theta/2 \end{aligned}$$

For the curve $r = b(1 - \cos \theta)$

$$\frac{dr}{d\theta} = b \sin \theta$$

$$\begin{aligned} \tan \phi_2 &= \frac{r}{b \sin \theta} = \frac{b(1 - \cos \theta)}{b \sin \theta} \\ &= \frac{2 \sin^2 \theta/2}{2 \sin \theta/2 \cos \theta/2} \\ &= \tan \theta/2 \end{aligned}$$

$$\begin{aligned} \tan \phi_1 \cdot \tan \phi_2 &= -\cot \theta/2 \cdot \tan \theta/2 \\ &= -1 \end{aligned}$$

∴ Given curves intersect orthogonally.

$$b. \quad r^n = a^n \cos n\theta, \quad r^n = b^n \sin n\theta.$$

→ For the curve $r^n = a^n \cos n\theta$

$$nr^{n-1} \frac{dr}{d\theta} = -a^n \sin n\theta \cdot n$$

$$\frac{dr}{d\theta} = \frac{-a^n \sin n\theta \cdot n}{nr^{n-1}}$$

$$\frac{dr}{d\theta} = \frac{-a^n \sin n\theta \cdot n}{r^{n-1}}$$

$$\tan \phi_1 = r \frac{d\theta}{dr} = -\frac{r \cdot r^{n-1}}{a^n \sin n\theta}$$

$$= \frac{-r^n}{a^n \sin n\theta} = \frac{-a^n \cos n\theta}{a^n \sin n\theta}$$

$$\tan \phi_1 = \frac{-\cos n\theta}{\sin n\theta}$$

$$\underline{\underline{\theta}}$$

For the curve $r^n = b^n \sin n\theta$.

$$nr^{n-1} \frac{dr}{d\theta} = b^n \cos n\theta \cdot n$$

$$\frac{dr}{d\theta} = \frac{b^n \cos n\theta \cdot n}{nr^{n-1}}$$

$$\tan \phi_2 = \frac{r \cdot nr^{n-1}}{b^n \cos n\theta}$$

$$= \frac{nr^n}{b^n \cos n\theta} = \frac{nb^n \sin n\theta}{b^n \cos n\theta}$$

$$\tan \phi_2 = n \tan n\theta$$

$$\therefore \tan \phi_1 \cdot \tan \phi_2 = -\cot n\theta \cdot \tan n\theta$$

$$= -1 //$$

$\therefore$ Given curves intersect orthogonally.

c. $r = a\theta$, $r = \frac{a}{\theta}$

For the curve $\frac{dr}{d\theta} = a$

$$\tan \phi_1 = \frac{a\theta}{a}$$

$$\tan \phi_1 = \underline{\underline{\theta}}$$

For the curve $\frac{dr}{d\theta} = \theta - \frac{a}{\theta^2}$

$$\tan \phi_2 = \frac{\theta}{\theta} \times \frac{\theta^2}{-a} = -\theta$$

$$\therefore \tan \phi_1 \cdot \tan \phi_2 = -\theta \cdot (-\theta)$$

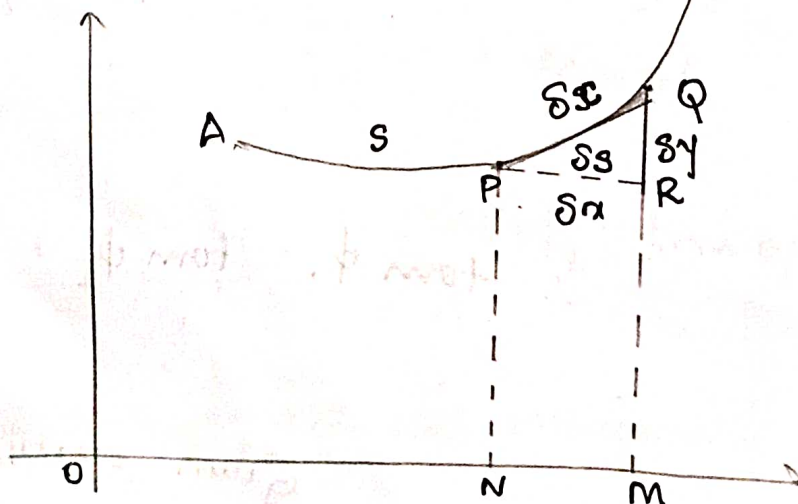
$$= \frac{\theta}{\theta} \times \frac{-a}{a} = -1$$

d. $r = \frac{a}{1+\cos\theta}$, $r = \frac{b}{1-\cos\theta}$

* imp.

> Derivative of arc:

For the cartesian curve $y = f(x)$, $\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$



Let $P(x, y)$, $Q(x + \delta x, y + \delta y)$ be 2 points on the curve AB. Let arc $AP = s$, arc $PQ = \delta s$ and chord $PQ = \delta c$. Draw $PN, QM \perp^{\text{to}} x\text{-axis}$ & $PR \perp QM$. From the right angled Δ^{e} PRQ ,

$$(PQ)^2 = (PR)^2 + (RQ)^2$$

$$(\delta c)^2 = (\delta x)^2 + (\delta y)^2$$

$$\div \delta x^2$$

$$\text{we have, } \left(\frac{\delta c}{\delta x} \right)^2 = 1 + \left(\frac{\delta y}{\delta x} \right)^2$$

$$\text{Now, } \left(\frac{\delta s}{\delta x} \right)^2 = \left(\frac{\delta s}{\delta c} \cdot \frac{\delta c}{\delta x} \right)^2 = \left(\frac{\delta s}{\delta c} \right)^2 \cdot \left[1 + \left(\frac{\delta y}{\delta x} \right)^2 \right]$$

Taking limit as $Q \rightarrow P$ (i.e. $\delta c \rightarrow 0$, $\delta x \rightarrow 0$)

$$\left[\frac{ds}{dx} \right]^2 = 1 + \left(\frac{dy}{dx} \right)^2$$

$$\therefore \boxed{\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx} \right)^2}} //$$

Note: For the curve $x = f(y)$

$$\therefore \boxed{\frac{ds}{dy} = \sqrt{1 + \left(\frac{dx}{dy} \right)^2}} //$$

$$d. \quad r = \frac{a}{1 + \cos \theta}, \quad \frac{b}{1 - \cos \theta}$$

$$\rightarrow \text{For the curve} \quad \frac{dr}{d\theta} = \frac{1 + \cos \theta \cdot 0 - a \sin \theta}{(1 + \cos \theta)^2} = \frac{-a \sin \theta}{(1 + \cos \theta)^2}$$

$$\tan \phi_1 = r \frac{d\theta}{dr} = \frac{r \cdot (1 + \cos \theta)^2}{a \sin \theta}$$

$$= \frac{a}{1 + \cos \theta} \cdot \frac{(1 + \cos \theta)^2}{a \sin \theta}$$

$$\rightarrow \frac{1 + \cos \theta}{\sin \theta} \cdot \frac{2 \cos \theta/2}{2 \sin \theta/2 \cos \theta/2} = \cot \theta/2$$

$$\rightarrow \frac{dr}{d\theta} = \frac{1 - \cos \theta \cdot 0 - b \sin \theta}{(1 - \cos \theta)^2} \cdot \frac{-b \sin \theta}{(1 - \cos \theta)^2}$$

$$\tan \phi_2 = \frac{r \cdot (1 - \cos \theta)^2}{-b \sin \theta} = \frac{b'}{1 - \cos \theta} \cdot \frac{(1 - \cos \theta)^2}{-b \sin \theta}$$

$$= \frac{1 - \cos \theta}{-\sin \theta}$$

$$= \frac{2 \sin \theta/2}{2 \sin \theta/2 \cos \theta/2} = \tan \theta/2$$

$$\tan \phi_1 \cdot \tan \phi_2 = -\cot \theta/2 \tan \theta/2 = \boxed{-1} //$$

Derivation continued:

If the equation of curve is in parametric form then

$$x = f(t), y = g(t)$$

$$\frac{ds}{dt} = \frac{ds}{dx} \cdot \frac{dx}{dt}$$

$$= \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \cdot \frac{dx}{dt}$$

$$\therefore \boxed{\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}}$$

$$\rightarrow \text{For the curve } r = f(\theta), \frac{ds}{d\theta} = \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}$$

Let $P(r, \theta)$ and $Q(r + \delta r, \theta + \delta \theta)$ be 2 neighbouring pts on the curve AB. Let arc AP = S , arc PQ = $S + \delta S$ and chord PQ = δS . Draw $PM \perp OQ$ then,
 $PM = r \delta \theta$

$$MQ = OQ - OM$$

$$= r + \delta r - r \cos \delta \theta$$

$$= \delta u + u(1 - \cos \delta \theta)$$

$$= \sigma_a + \lambda \frac{2 \sin^2 \frac{\sigma_0}{2}}$$

From the right angled $\triangle$ let

$$(PQ)^2 = (PM)^2 + (MQ)^2$$

$$(\delta c)^2 = r^2 \sin^2 \delta \theta +$$

$$\left(S r + 2 \lambda \sin^2 \frac{\delta \theta}{2} \right)^2$$

$$\div (50)^2$$

$$\left(\frac{\partial c}{\partial \theta}\right)^2 = r^2 \left(\frac{\sin \theta}{\theta}\right)^2 + \left(\frac{\partial r}{\partial \theta} + \frac{2r \sin^2 \frac{\theta}{2}}{\theta}\right)^2$$

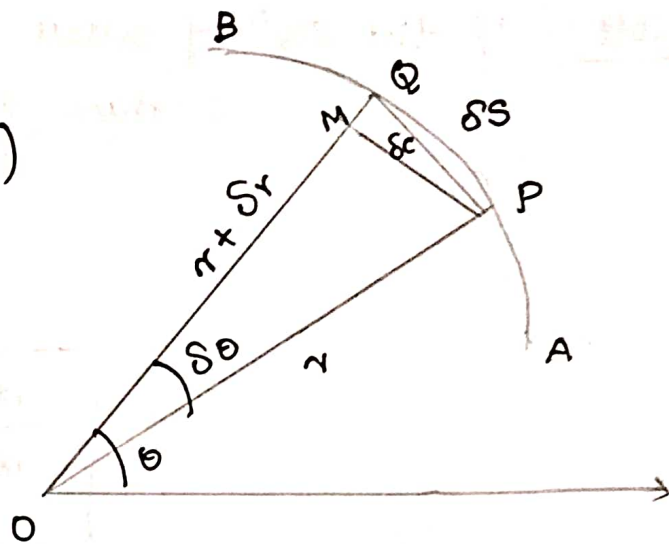
$$\left(\frac{\delta s}{\delta \theta}\right)^2 = \left(\frac{\delta s}{\delta c} \cdot \frac{\delta c}{\delta \theta}\right)^2 = \left(\frac{\delta s}{\delta c}\right)^2 \left[r^2 \left(\frac{\sin \theta}{\delta \theta}\right)^2 + \left(\frac{\delta \lambda}{\delta \theta} + \frac{2\lambda \sin^2 \theta}{2} \frac{\delta \theta}{\delta \theta}\right)^2 \right]$$

$$\left(\frac{\partial s}{\partial \theta}\right)^2 = \left(\frac{\partial s}{\partial c}\right)^2 \left[r^2 \left(\frac{\sin \delta \theta}{\delta \theta} \right)^2 + \left(\frac{\partial a}{\partial \theta} + a \frac{\sin \delta \theta}{\delta \theta} \right)^2 \right]$$

Taking limit as $Q \rightarrow P$

$$\left(\frac{ds}{d\theta}\right)^2 = l^2 \left[u^2 + \left(\frac{du}{d\theta} + r \cdot O \cdot l \right)^2 \right]$$

$$\therefore \frac{ds}{d\theta} = \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}$$



Note: If the eqⁿ of curve $\theta = f(r)$

$$\text{then } \frac{ds}{dr} = \frac{ds}{d\theta} \cdot \frac{d\theta}{dr}$$

$$= \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} \cdot \frac{d\theta}{dr}$$

$$\boxed{\frac{ds}{dr} = \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2 + 1}} \quad *$$

* Find $\frac{ds}{dx}$ for $xy^2 = x^3$

Sol: $2xy \frac{dy}{dx} = 3x^2$

$$\frac{dy}{dx} = \frac{3x^2}{2xy}$$

$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$= \sqrt{1 + \frac{9x^4}{4x^2y^2}}$$

$$= \sqrt{1 + \frac{9x^2}{4y^2}} \quad \begin{matrix} \xrightarrow{4a^2y^2} a^2y^2 \\ \xrightarrow{a^2y^2} x^3 \end{matrix}$$

$$= \sqrt{1 + \frac{9x}{4a}}$$

$$= \sqrt{\frac{4+9}{4}}$$

* Find $\frac{ds}{d\theta}$ for the curves

$$1) r = a (1 - \cos \theta)$$

Differentiating w.r.t θ , $\frac{dr}{d\theta} = a \sin \theta$

$$\begin{aligned} \frac{ds}{d\theta} &= \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} \\ &= \sqrt{a^2(1 - \cos \theta)^2 + a^2 \sin^2 \theta} \\ &= \sqrt{a^2 [1 + \cos^2 \theta - 2\cos \theta + \sin^2 \theta]} \\ &= a \sqrt{2(1 - \cos \theta)} \\ &= a \sqrt{2} \sqrt{2 \sin^2 \theta / 2} \end{aligned}$$

$$2) r^2 = a^2 \cos 2\theta.$$

$$2r \frac{dr}{d\theta} = -2a^2 \sin 2\theta.$$

$$\begin{aligned} \frac{ds}{d\theta} &= \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} \\ &= \sqrt{r^2 + \left(\frac{-a^2 \sin 2\theta}{2r}\right)^2} \\ &= \sqrt{a^2 \cos 2\theta + \left(\frac{-a^2 \sin 2\theta}{2r}\right)^2} \\ &= \sqrt{a^2 \cos 2\theta + \frac{a^4 \sin^2 2\theta}{4r^2}} \end{aligned}$$

$$= \sqrt{\frac{4r^2 a^2 \cos 2\theta + a^4 \sin^2 2\theta}{4r^2}}$$

$$\begin{aligned} &= \sqrt{\frac{a^4 \cos^2 2\theta + a^4 \sin^2 2\theta}{a^2 \cos 2\theta}} = a \sqrt{\frac{1}{\cos 2\theta}} \\ &= a \sqrt{\sec 2\theta} \end{aligned}$$

• Find $\frac{ds}{d\theta}$, $\frac{ds}{dx}$, $\frac{ds}{dy}$, $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$

Sol: $\frac{dx}{d\theta} = a(1 - \cos\theta)$

$\frac{dy}{d\theta} = a \sin\theta$

$$\frac{ds}{d\theta} = \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} = \sqrt{a^2(1 - \cos\theta)^2 + a^2 \sin^2\theta}$$

$$\frac{ds}{d\theta} = a \sqrt{a^2(1 - \cos\theta)^2 + a^2 \sin^2\theta}$$

$$= a \sqrt{\frac{(1 - \cos\theta)^2}{(a-b)^2} + \sin^2\theta}$$

$$= a \sqrt{2(1 - \cos\theta)}$$

$$= a\sqrt{2} \sqrt{2\sin^2\theta/2}$$

$$= 2a \sin\theta/2 //$$

$$\frac{ds}{dx} = \frac{\frac{ds}{d\theta}}{\frac{dx}{d\theta}} = \operatorname{cosec} \theta/2 //$$

$$\frac{ds}{dy} = \frac{\frac{ds}{d\theta}}{\frac{dy}{d\theta}} = \sec \theta/2 //$$

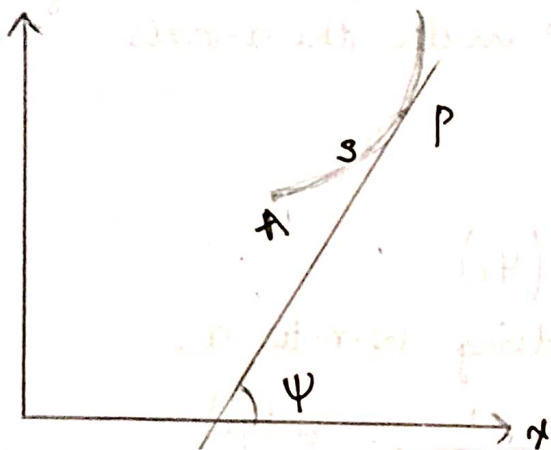
• Find $\frac{ds}{d\theta}$ for the curve $r = a(1 + \cos\theta)$.

$$\rightarrow \frac{ds}{d\theta} = \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}$$

$$= \frac{dr}{d\theta} = -a \sin \theta.$$

$$\begin{aligned} \therefore \frac{ds}{d\theta} &= \sqrt{a^2(1+\cos\theta)^2 + (-a^2\sin^2\theta)} \\ &= \sqrt{a^2(1+\cos\theta)^2 + a^2\sin^2\theta} \\ &= a\sqrt{2(1+\cos\theta)} \\ &= a\sqrt{2} \sqrt{2\cos^2\theta/2} \therefore = \underline{2a\cos\theta/2} \end{aligned}$$

> Curvature of a curve at a given point is the measure of rate of change of bending of curve at that point.



Let 'P' be a point on the curve $y = f(x)$ at a length s from a fixed point A on it.

Let tangent at P makes an angle ψ with the direction of x -axis as a point P moves along the curve both s and ψ vary. The rate of change of ψ w.r.t s

$\frac{d\psi}{ds}$ is called the curvature of the curve P.

The reciprocal of the curvature at P is called the radius of curvature at P and is denoted by ρ

$$\therefore \boxed{\rho = \frac{ds}{d\psi}} //$$

→ Radius of curvature for cartesian curve $y = f(x)$ is

given by

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

previous
same diagram refer

Let $y = f(x)$ be the equation of cartesian curve.

Let $P(x, y)$ be a point on the curve such that arc $AP = s$. Let ψ be the angle made by the tangent P with the x -axis. Then we know that

$$\tan \psi = \frac{dy}{dx} = y_1$$

$$\text{or } \psi = \tan^{-1}(y_1)$$

Differentiating w.r. to x ,

$$\frac{d\psi}{dx} = \frac{1}{1 + y_1^2} \cdot \frac{d(y_1)}{dx}$$

$$\frac{d\psi}{dx} = \frac{y_2}{1 + y_1^2}$$

Now,

$$\rho = \frac{ds}{d\psi} = \frac{ds}{dx} \cdot \frac{dx}{d\psi}$$

$$\rho = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \cdot \frac{1 + y_1^2}{y_2}$$

$$= \sqrt{1 + y_1^2} \cdot \frac{1 + y_1^2}{y_2}$$

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

* Parametric derivation for Cartesian curve:

Let $x = f(t)$, $y = \phi(t)$

We have for Cartesian curve $y = f(x)$, $\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} \rightarrow \textcircled{*}$

Denoting differentiation w.r.t t by dashes,

$$y_1 = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{y'}{x'}$$

$$y_2 = \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

$$y_2 = \frac{d}{dx} \left(\frac{y'}{x'} \right)$$

$$= \frac{d}{dt} \left(\frac{y'}{x'} \right) \cdot \frac{dx}{dt}$$

$$= \frac{\frac{x'y'' - y'x''}{x'^2}}{x'}$$

$$y_2 = \frac{x'y'' - y'x''}{(x')^3}$$

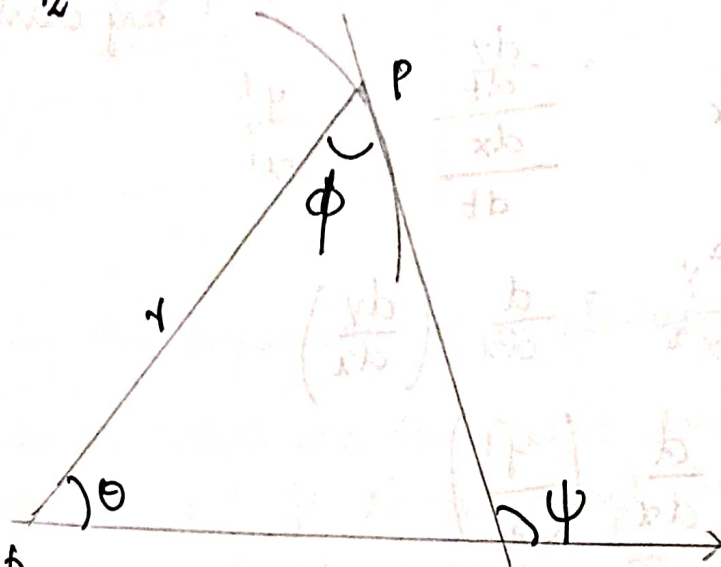
$$\begin{aligned} \therefore \textcircled{*} &\Rightarrow \rho = \left(1 + \left(\frac{y'}{x'} \right)^2 \right)^{3/2} \times \frac{(x')^3}{x'y'' - y'x''} \\ &= \frac{(x'^2 + y'^2)^{3/2}}{(x')^3} \times \frac{(x')^3}{x'y'' - y'x''} \end{aligned}$$

$$\therefore \boxed{\rho = \frac{(x'^2 + y'^2)^{3/2}}{x'y'' - y'x''}}$$

* Radius of curvature for polar curve $r = f(\theta)$ is given by

$$\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - rr_2}$$

Proof:



Let $OP = r$ with a radius vector and ϕ be the angle made by the radius vector with the tangent at $P(r, \theta)$.

Let ψ be the angle made by the tangent at P with initial line. From figure $\psi = \theta + \phi$.

Differentiating w.r.t θ ,

$$\begin{aligned} \frac{d\psi}{ds} &= \frac{d\theta}{ds} + \frac{d\phi}{ds} \\ &= \frac{d\theta}{ds} + \frac{d\phi}{d\theta} \cdot \frac{d\theta}{ds} \end{aligned}$$

$$\frac{1}{\rho} = \frac{d\theta}{ds} \left[1 + \frac{d\phi}{d\theta} \right] \longrightarrow (1)$$

We know that $\frac{ds}{d\theta} = \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}$

$$\frac{ds}{d\theta} = \sqrt{r^2 + r_1^2}$$

$$\frac{d\theta}{ds} = \frac{1}{\sqrt{r^2 + r_1^2}} \longrightarrow (2)$$

we have, $\tan \phi = \frac{r \frac{d\theta}{dr}}$

$$\tan \phi = \frac{r}{\lambda_1}$$

$$\phi = \tan^{-1} \left(\frac{r}{\lambda_1} \right)$$

$$\frac{d\phi}{d\theta} = \frac{1}{1 + \left(\frac{r}{\lambda_1} \right)^2} \cdot \frac{r_1^2 - r r_2}{r_1^2}$$

$$\frac{d\phi}{d\theta} = \frac{r_1^2 - r r_2}{r_1^2 + r^2} \rightarrow (3)$$

$$(1) \Rightarrow \frac{1}{\rho} = \frac{1}{\sqrt{r^2 + r_1^2}} \left[1 + \frac{r_1^2 - r r_2}{r_1^2 + r^2} \right]$$

$$\frac{1}{\rho} = \frac{1}{\sqrt{r^2 + r_1^2}} \left[\frac{r_1^2 + r^2 + r_1^2 - r r_2}{r_1^2 + r^2} \right]$$

$$\frac{1}{\rho} = \frac{r^2 + 2r_1^2 - r r_2}{(r^2 + r_1^2)^{3/2}}$$

$$\therefore \boxed{\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - r r_2}}$$

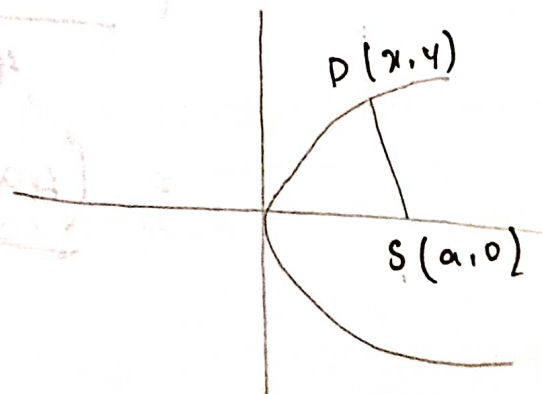
- Let ρ be the radius of curvature at any pt P on the parabola $y^2 = 4ax$ and S be its focus also show that ρ varies as $(SP)^3$.

$$\rightarrow y^2 = 4ax$$

$$2y y_1 = 4a$$

$$y_1 = \frac{2a}{y}$$

$$\boxed{y_1 = \frac{2a}{y}}$$



$$\gamma_2 = \frac{-2a}{\gamma^2} \cdot \gamma_1$$

$$\gamma_2 = -\frac{2a}{\gamma^2} \cdot \frac{2a}{\gamma}$$

$$\therefore \left[\gamma_2 = \frac{-4a^2}{\gamma^3} \right]$$

$$\rightarrow \int = \frac{\left(1 + \gamma_1^2 \right)^{3/2}}{\gamma_2}$$

$$= \frac{\left(1 + \left(\frac{2a}{\gamma} \right)^2 \right)^{3/2}}{\frac{-4a^2}{\gamma^3}}$$

$$= \frac{\left(\frac{\gamma^2 + 4a^2}{\gamma^2} \right)^{3/2}}{\frac{-4a^2}{\gamma^3}}$$

$$= \frac{(\gamma^2 + 4a^2)^{3/2}}{\cancel{\gamma^3}} \times \frac{\gamma^3}{-4a^2}$$

$$= \frac{(4a^2 + 4a^2)^{3/2}}{4a^2}$$

$$= \frac{(4a(\gamma + a))^{3/2}}{-4a^2}$$

$$= \frac{(4a)^{3/2} (x+a)^{3/2}}{-4a^2}$$

$$= \frac{4 \cdot 4^{1/2}}{-4} \frac{a^{3/2}}{a^2} (x+a)^{3/2}$$

$$= \frac{-2}{\sqrt{a}} (x+a)^{3/2}$$

$$p^2 = \frac{4}{a} (x+a)^3 \rightarrow (*)$$

$$\text{Now } SP = \sqrt{(x-a)^2 + y^2}$$

$$= \sqrt{x^2 + a^2 - 2xa + 4ax}$$

$$SP = x+a \rightarrow \textcircled{1}$$

From $\textcircled{1}$,

$$p^2 = \frac{4}{a} (x+a)^3$$

$$p^2 = \frac{4}{a} (SP)^3$$

$$\Rightarrow p^2 \propto (SP)^3 //$$

• Show that the radius of curvature of the cardio is
 $r = a(1 - \cos \theta)$ varies as $\sqrt{x}$

$$\rightarrow \frac{dr}{d\theta} = r_1 = a \sin \theta$$

$$\frac{dr_1}{d\theta} r_2 = a \cos \theta$$

$$p = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - r r_2}$$

$$= \frac{(a^2(1 - \cos \theta)^2 + a^2 \sin^2 \theta)^{3/2}}{(a^2(1 - \cos \theta)^2 + 2a^2 \sin^2 \theta - a(1 - \cos \theta) - a(1 - \cos \theta)a \cos \theta)}$$

$$\rho = \frac{\left[a^2 [1 + \cos^2 \theta - 2\cos \theta + \sin^2 \theta] \right]^{3/2}}{a^2 [1 + \cos^2 \theta - 2\cos \theta + 2\sin^2 \theta - \cos \theta + \cos^2 \theta]}$$

$$\rho = \frac{[a^3 \cdot 2[1 - \cos \theta]]^{3/2}}{a^2 [3 - 3\cos \theta]}$$

$$\rho = \frac{a^3 (2(1 - \cos \theta))^{3/2}}{3a^2 (1 - \cos \theta)}$$

$$= \frac{2^{3/2} a (1 - \cos \theta)^{3/2}}{3 (1 - \cos \theta)} \cdot \frac{(1 - \cos \theta)^{1/2} (1 - \cos \theta)}{(1 - \cos \theta)}$$

$$= \frac{a \cdot 2^{3/2}}{3} \sqrt{1 - \cos \theta}$$

$$= \frac{2^{3/2}}{3} \sqrt{a} \sqrt{a(1 - \cos \theta)}$$

$$= \frac{2^{3/2}}{3} \sqrt{a} \sqrt{r}$$

$$\therefore \rho \propto \sqrt{x}$$

Hence proved...

- If ρ_1 and ρ_2 are radius of curvature at the extremities of any chord of cardioid $r = a(1 + \cos \theta)$ which passes through the pole, show that $\rho_1^2 + \rho_2^2 = \frac{16a^2}{9}$ //

differentiating $r = a(1 + \cos \theta)$

$$r_1 = -a \sin \theta$$

$$r_2 = -a \cos \theta$$

$$\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - rr_2}$$

$$\rho = \frac{(a^2(1+\cos\theta)^2 + (-a\sin\theta)^2)^{3/2}}{a^2(1+\cos\theta)^2 + 2(-a\sin\theta)^2 - a(1+\cos\theta)(-a\cos\theta)}$$

$$= \frac{(a^2(1+\cos^2\theta + 2\cos\theta + \sin^2\theta))^{3/2}}{a^2(1+\cos\theta)^2 + 2a^2\sin^2\theta + a(1+\cos\theta)(a\cos\theta)}$$

$$= \frac{(a^2(1+\cos^2\theta + 2\cos\theta + \sin^2\theta))^{3/2}}{a^2(1+\cos\theta)^2 + 2a^2\sin^2\theta + a(1+\cos\theta)(a\cos\theta)}$$

$$= \frac{(a^2(1+\cos^2\theta + 2\cos\theta + \sin^2\theta))^{3/2}}{a^2(1+\cos\theta)^2 + 2a^2\sin^2\theta + a(1+\cos\theta)(a\cos\theta)}$$

$$a^2 \left[(1+\cos\theta)^2 + 2\sin^2\theta + \cos\theta + \cos^2\theta \right]$$

$$= \frac{a^3 (2(1+\cos\theta))^{3/2}}{3a^2 [3(1+\cos\theta)]}$$

$$= \frac{a^3 (2(1+\cos\theta))^{3/2}}{3a^2 [3(1+\cos\theta)]}$$

$$= \frac{a^3 2^{3/2} (1+\cos\theta)^{3/2}}{3 (1+\cos\theta)}$$

$$= \frac{a^3 2^{3/2}}{3} \sqrt{1+\cos\theta}$$

$$= \frac{a^3 2^{3/2}}{3} \sqrt{2\cos^2\theta/2}$$

$$= \frac{a^3 2^{3/2}}{3} \sqrt{2} \cdot \cos\theta/2$$

$$\therefore \boxed{\rho = \frac{4}{3} a \cos\theta/2}$$

$$\therefore \rho_1 = \frac{4}{3} a \cos\theta/2$$

$$\rho_2 = \frac{4}{3} a \cos\left(\frac{\pi+\theta}{2}\right)$$

$$= \frac{4}{3} a (-\sin\theta/2)$$

$$\therefore f_1^2 + f_2^2 = \frac{16}{9} a^2 \cos^2 \frac{\theta}{2} + \frac{16}{9} a^2 \sin^2 \frac{\theta}{2}$$

$$= \frac{16a^2}{9} //$$

• Prove that radius of curvature of a circle is constant

Sol: $x = r \cos \theta$ $y = r \sin \theta$

$$\rho = \frac{(x'^2 + y'^2)^{3/2}}{x'y'' - y'x''}$$

$$x' = -r \sin \theta \quad y' = r \cos \theta$$

$$y'' = -r \cos \theta \quad x'' = -r \sin \theta$$

$$\begin{aligned} \rho &= \frac{r^2 \sin^2 \theta + r^2 \cos^2 \theta}{-r \sin \theta (-r \sin \theta) - r \cos \theta (-r \cos \theta)} \\ &= \frac{(r^2 \sin^2 \theta + r^2 \cos^2 \theta)^{3/2}}{r^2 \sin^2 \theta + r^2 \cos^2 \theta} \\ &= \frac{(r^2)^{3/2}}{r^2} \end{aligned}$$

$$\therefore \boxed{\rho = r}$$

Find ρ at the point $t=1$ on the curve $x = at^2$, $y = 2at$

Sol: $x' = 2at$

$$x'' = 2a$$

$$y' = 2a$$

$$y'' = 0$$

$$\begin{aligned} f &= \frac{(x'^2 + y'^2)^{3/2}}{x'y'' - y'x''} \\ &= \frac{((2at)^2 + (2a)^2)^{3/2}}{2at \cdot 0 - 2a \cdot 2a} \\ &= \frac{(4a^2t^2 + 4a^2)^{3/2}}{-4a^2} \end{aligned}$$

$$\text{at } t=1 \quad f = \frac{(8a^2)^{3/2}}{4a^2} = \frac{a^3 8^{3/2}}{4a^2} = \frac{a^3 8^2 8^{1/2}}{4a^2}$$

$$= -4\sqrt{2}a$$

$$\therefore \boxed{f = 4\sqrt{2}a}$$

• Show that the radius of curvature at the point $\left(\frac{3a}{2}, \frac{3a}{2}\right)$ of $x^3 + y^3 = 3axy$ is $\frac{3a}{8\sqrt{2}}$.

$$\rightarrow x^3 + y^3 = 3axy$$

$$3x^2 + 3y^2 \cdot \frac{dy}{dx} = 3a[x_1 + y]$$

$$\begin{aligned} 3y^2 \frac{dy}{dx} &= \frac{3a[x_1 + y]}{3x^2} \\ \frac{dy}{dx} &= \frac{a[x_1 + y]}{xy^2} \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{omit}$$

$$3x^2 + 3y^2 y_1 = 3a[x_1 + y] y_1 = ax$$

$$x^2 + y^2 y_1 = a[x_1 + y] y_1 = a[x_1 + y]$$

$$x^2 + y^2 y_1 = ax_1 + ay$$

$$x^2 - ax_1 = ax_1 - y^2 y_1$$

$$x^2 - ax_1 = y_1 (ax_1 - y^2)$$

$$y^2 y_1 = a[x_1 + y] - x^2$$

$$y_1 = \frac{a[x_1 + y] - x^2}{y^2}$$

$$y_1 = \frac{ax_1 + ay - x^2}{y^2}$$

$$\frac{y_1 - ax_1}{y^2} = \frac{ay - x^2}{y^2}$$

$$y_1 = \frac{ay - x^2}{y^2 - ax}$$

$$y_2 = \frac{(y^2 - ax)(ay_1 - 2x) - (ay - x^2)(2yy' - a)}{(y^2 - ax)^2}$$

$$\text{at } \left(\frac{3a}{2}, \frac{3a}{2} \right), \quad y_1 = \frac{\frac{3a^2}{2} - \frac{9a^2}{4}}{\frac{9a^2}{4} - \frac{3a^2}{2}}$$

$$\frac{\frac{12a^2 - 18a^2 - 1}{8}}{\frac{18a^2 - 12a^2}{8}} \rightarrow = -1$$

$$y_2 = \frac{\left(\frac{9a^2}{4} - \frac{3a^2}{2} \right) \left(-a - 2 \times \frac{3a}{2} \right) - \left(\frac{3a^2}{2} - \frac{9a^2}{4} \right) \left(2 \times \frac{3a}{2} (-1) - a \right)}{\left(\frac{9a^2}{4} - \frac{3a^2}{2} \right)^2}$$

$$y_2 = \frac{-8a}{\frac{9a^2}{4} - \frac{3a^2}{2}} = \frac{-8a}{\frac{9a^2 - 6a^2}{4}} = \frac{-8a \times 4}{3a^2}$$

$$= -\frac{32}{3a}$$

> Rolle's theorem:

If a function $f(x)$ is continuous

$$f = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

$$= \frac{(1+1)^{3/2} \times 3a}{32}$$

$$= \frac{2^{3/2} \times 3a}{32}$$

$$f = \frac{-2 \times \sqrt{2} \times 3a}{32} = \frac{-\sqrt{2} \times 3a}{16} = \frac{-\sqrt{2} \times 3a}{8\sqrt{2} \sqrt{2}} = \frac{3a}{8\sqrt{2}}$$

$$\therefore \boxed{f = \frac{3a}{8\sqrt{2}}}$$

Rolle's Theorem:

- 1) A function $f(x)$ is continuous in (a, b) .
- 2) $f(x)$ exists for every value of x in (a, b) .
- 3) $f(a) = f(b)$ then there is atleast one value 'c' in (a, b) such that $f'(c) = 0$.

Lagrange's MVT:

- If
- (1) $f(x)$ is continuous $[a, b]$.
 - (2) $f(x)$ is differentiable in (a, b) then there exists atleast one point c in open interval (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Steps:

(i) (i) $f(x) = \log x$ in $[e, e^2]$

(ii) $f'(x) = \frac{1}{x}$ exists (e, e^2)

(iii) $f(x)$ is continuous in $[e, e^2]$

($\because$ Every differentiable function is continuous)

$\therefore$ By LMVT,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\frac{1}{c} = \frac{\log e^2 - \log e}{e^2 - e}$$

$$c = \frac{e^2 - e}{\log \left(\frac{e^2}{e} \right)}$$

$$\therefore c = e^2 - e \in [e, e^2]$$

(2) $f(x) = \tan^{-1} x$ in $[0, 1]$

$$f'(x) = \frac{1}{1+x^2} \text{ exists } (0, 1)$$

$f(x)$ is continuous in $[0, 1]$

$\therefore$ By LMVT,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$= \frac{\tan^{-1}(1) - \tan^{-1}(0)}{1 - 0}$$

$$\frac{1}{1+c^2} = \frac{\pi}{4}$$

$$\therefore e = \sqrt{\frac{4}{\pi}} \quad \frac{1}{1+c^2} = \frac{4}{\pi} \frac{\pi}{4}$$

$$1+c^2 = \frac{4}{\pi}$$

$$c^2 = \frac{4}{\pi} - 1$$

$$\therefore \boxed{c = \pm \sqrt{\frac{4}{\pi} - 1}}$$

$$c = 0.52 \in [0, 1]$$

$$3) f(x) = \sin^{-1} x \text{ in } [0, 1]$$

$$f'(x) = \frac{1}{\sqrt{1-x^2}} \text{ exists } (0, 1)$$

$f(x)$ is continuous in $[0, 1]$

$\therefore$ By LMVT,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\frac{1}{\sqrt{1-x^2}} = \frac{\sin^{-1}(1) - \sin^{-1}(0)}{1 - 0}$$

$$\frac{1}{\sqrt{1-x^2}} = \frac{\frac{\pi}{2} - 0}{1}$$

$$\frac{1}{\sqrt{1-x^2}} = \frac{\pi}{2}$$

$$\sqrt{1-x^2} = \frac{2}{\pi}$$

$$1-x^2 = \frac{4}{\pi^2}$$

$$x^2 = 1 - \frac{4}{\pi^2}$$

Say, on B.S

$$1 - c^2 = \frac{4}{\pi^2}$$

$$c^2 = 1 - \frac{4}{\pi^2}$$

$$\therefore \boxed{c = \pm \sqrt{1 - \frac{4}{\pi^2}}}$$

Cauchy's Mean Value Theorem

If two functions $f(x)$ and $g(x)$ satisfy the following condition:

- ✓ $f(x)$ and $g(x)$ are continuous $[a, b]$
- ✓ $f(x)$ and $g(x)$ are differentiable (a, b) .
- ✓ $g(x) \neq 0$ for every x in (a, b)

Then there exist atleast one point ' c ' in (a, b) (i.e)

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Proof: Let us define

$$f(x) = f(x) + Ag(x)$$

where A is const such that

$$f(a) = f(b)$$

thus

$$f(a) + Ag(a) = f(b) + Ag(b)$$

$$A(g(a) - g(b)) = f(b) - f(a)$$

$$A = \frac{f(b) - f(a)}{g(b) - g(a)}, \quad g(b) - g(a) \neq 0.$$

Now, F satisfies the conditions Rolle's theorem $[a, b]$.

$\therefore$ There is atleast one value $c \in (a, b)$ such that $F'(c) = 0$. We have

$$\text{such that } F'(x) = f'(x) + Ag'(x)$$

$$\therefore F'(c) = 0 \Rightarrow f'(c) + Ag'(c) = 0$$

$$f'(c) = -Ag'(c)$$

$$\therefore \boxed{\frac{f'(c)}{g'(c)} = -A}$$

7. Verify Cauchy's mean value theorem for the functions:

1. $f(x) = x^3$, $g(x) = x^2$ in $[1, 2]$.

Sol: (i) $f'(x) = 3x^2$, $g'(x) = 2x$ exists in $(1, 2)$.

(ii) $f(x)$ and $g(x)$ are continuous in $[a, b]$.

(iii) $g'(x) \neq 0 \quad \forall x$ in $(1, 2)$.

$\therefore$ By C.M.V.T there exists one value in $(1, 2)$ such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

$$\frac{3c^2}{2c} = \frac{8-1}{4-1}$$

$$\frac{3}{2}c = \frac{7}{3}$$

$$c = \frac{7}{3} \times \frac{2}{3}$$

$$\boxed{c = 1.55 \in (1, 2)}$$

2. $f(x) = e^x$, $g(x) = e^{-x}$
in $[a, b]$,
 $b > a > 0$

Sol: (i) $f'(x) = e^x$, $g'(x) = -e^{-x}$ exists in (a, b)

(ii) $f(x)$ and $g(x)$ are continuous in $[a, b]$.

(iii) $g'(x) \neq 0 \quad \forall x$ in (a, b)

$$\therefore \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

$$\frac{e^c}{-e^{-c}} = \frac{e^b - e^a}{e^{-b} - e^{-a}}$$

$$\frac{e^c}{-1/e^c} = \frac{e^b - e^a}{1/e^b - 1/e^a}$$

$$= -e^{2c} = \frac{e^b - e^a}{e^a - e^b}$$

$$= -e^{2c} = \frac{e^b - e^a}{e^a - e^b} \times e^{a+b}$$

$$= -e^{2c} = -1 \times e^{a+b}$$

$$= e^{2c} = e^{a+b}$$

$$= 2c = a+b$$

$$= \boxed{c = \frac{a+b}{2} \in (a,b)}$$

3. $f(x) = \frac{1}{x^2}$, $g(x) = \frac{1}{x}$ in $[a,b]$, $b > a > 0$

sol: (i) $f'(x) = -2x^{-3} = -\frac{2}{x^3}$, $g'(x) = -\frac{1}{x^2}$ exists in (a,b)

(ii) $f(x)$ and $g(x)$ are continuous in $[a,b]$

(iii) $g'(x) \neq 0 \forall x$ in (a,b) .

$$\therefore \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

$$\frac{-2c^{-3}}{-c^{-2}} = \frac{\frac{1}{b^2} - \frac{1}{a^2}}{\frac{1}{b} - \frac{1}{a}}$$

$$\frac{2}{c} = \frac{\frac{a^2 - b^2}{a^2 b^2}}{\frac{a - b}{ab}}$$

$$\frac{2}{c} = \frac{a^2 - b^2}{(ab)(a-b)} = \frac{(a+b)(a-b)}{(ab)(a-b)} \therefore \boxed{c = \frac{2ab}{a+b}}$$

omit $\therefore \boxed{c = \frac{a^2 b - ab^2}{a^2 - b^2} \in (a,b)}$ $\boxed{c = \frac{2(a^2 b - ab^2)}{a^2 - b^2}}$

4. $f(x) = \log x$, $g(x) = \frac{1}{x}$ in $[1, 2]$.

- Sol: (i) $f'(x) = \frac{1}{x}$, $g'(x) = -x^{-2}$ exists in (a, b) .
 (ii) $f(x)$ and $g(x)$ are continuous in $[1, 2]$
 (iii) $g'(x) \neq 0$ $\forall x$ is $(1, 2)$

$$\therefore \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)} \quad \text{L'Hôpital's Rule}$$

$$\frac{1/x}{-1/x^2} = \frac{\log 2 - \log 1}{1/e - 1} = \frac{1-0}{\frac{1}{e}-1}$$

$$-c = \frac{2(\log 2 - \log 1)}{1-2}$$

$$-c = \frac{1/2(\log 2 - \log 1)}{1}$$

$$= \frac{e}{1-e}$$

$$\therefore \sqrt{c = \frac{e}{e-1}}$$

$$\therefore \boxed{c = \frac{2 \log 2}{2 \log 2 - 1}}$$

> Taylor's series:

Taylor's series function expansion of a function $f(x)$ at $x=a$.

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) +$$

$$\frac{(x-a)^3}{3!} f'''(a) + \frac{(x-a)^4}{4!} f^{(4)}(a)$$

> Maclaurin's Series :

Put $a=0$ in Taylor's series expansion we get

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0)$$

- (i) Obtain Taylor's series of $\sin(x)$ in period of $(x - \frac{\pi}{2})$ upto terms containing $(x - \frac{\pi}{2})^4$

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \frac{(x-a)^4}{4!} f^{(4)}(a)$$

$$\begin{aligned} f(x) &= \sin x & f(\pi/2) &= 1 \\ f'(x) &= \cos x & f'(\pi/2) &= 0 \\ f''(x) &= -\sin x & f''(\pi/2) &= -1 \\ f'''(x) &= -\cos x & f'''(\pi/2) &= 0 \\ f^{(4)}(x) &= \sin x & f^{(4)}(\pi/2) &= 1 \end{aligned}$$

$$\sin x = 1 + (x - \pi/2) \cdot 0 + \frac{(x - \pi/2)^2}{2!} (-1) + \frac{(x - \pi/2)^3}{3!} \cdot 0 + \frac{(x - \pi/2)^4}{4!} \cdot 1$$

$$\therefore \sin x = 1 - \frac{(x - \pi/2)^2}{2!} + \frac{(x - \pi/2)^4}{4!} + \dots$$

(ii) Obtain Taylor's expansion of $\log x$ at $x=1$ upto 3 non-vanishing terms.

$$\begin{aligned} f(x) &= \log x & f(1) &= 0 \\ f'(x) &= \frac{1}{x} & f'(1) &= 1 \end{aligned}$$

$$f''(x) = -x^{-2} \quad f''(1) = -1$$

$$f'''(x) = -2x^{-3} \quad f'''(1) = -2$$

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \frac{(x-a)^4}{4!} f^{(4)}(a) + \dots$$

$$f(x) = \sin x \quad f(\pi/2) = 1$$

$$f'(x) = \cos x \quad f'(\pi/2) = 0$$

$$f''(x) = -\sin x \quad f''(\pi/2) = -1$$

$$f'''(x) = -\cos x \quad f'''(\pi/2) = 0$$

$$f^{(4)}(x) = \sin x \quad f^{(4)}(\pi/2) = 1$$

$$\sin x = 1 + (x - \pi/2) \cdot 0 + \frac{(x - \pi/2)^2}{2!} (-1) + \frac{(x - \pi/2)^3}{3!} \cdot 0 + \frac{(x - \pi/2)^4}{4!} \cdot 1 + \dots$$

$$\therefore \sin x = 1 - \frac{(x - \pi/2)^2}{2!} + \frac{(x - \pi/2)^4}{4!} - \dots$$

(ii) Obtain Taylor's expansion of $\log x$ at $x=1$ upto 3 non-vanishing terms.

$$f(x) = \log x \quad f(1) = 0$$

$$f'(x) = \frac{1}{x} \quad f'(1) = 1$$

$$f''(x) = -x^{-2} \quad f''(1) = -1$$

$$f'''(x) = -2x^{-3} \quad f'''(1) = -2$$

$$\therefore \log x = \cancel{0} + (x-1) \cdot 1 + \frac{(x-1)^2}{2!} (-1) + \frac{(x-1)^3}{3!} (2) \dots$$

$$= (x-1) \mp \frac{(x-1)^2}{2!} + \frac{(x-1)^3}{3!} \cdot 2 + \dots$$

(iii) e^x

$$f(x) = e^x$$

$$f(x) = x f'(x) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0)$$

$$\begin{aligned} f(x) &= e^x & f(0) &= 1 \\ f'(x) &= e^x & f'(0) &= 1 \\ f''(x) &= e^x & f''(0) &= 1 \\ f'''(x) &= e^x & f'''(0) &= 1 \end{aligned}$$

$$\therefore e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

(iv) $\sin x$

$$\begin{aligned} f(x) &= \sin x & f(0) &= 0 \\ f'(x) &= \cos x & f'(0) &= 1 \\ f''(x) &= -\sin x & f''(0) &= 0 \\ f'''(x) &= -\cos x & f'''(0) &= -1 \end{aligned}$$

$$\therefore \sin x = 0 + \sin x \cdot 1 + \frac{x^2}{2!} \cdot 0 + \frac{x^3}{3!} (-1) + \frac{x^4}{4!} (0) + \frac{x^5}{5!} (1)$$

$$\sin x = x \mp \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$$

(v) $\cos x$

$$f(x) = \cos x$$

$$f(0) = 1$$

$$f'(x) = -\sin x$$

$$f'(0) = 0$$

$$f''(x) = -\cos x$$

$$f''(0) = -1$$

$$f'''(x) = \sin x$$

$$f'''(0) = 0$$

$$f^{(4)}(x) = \cos x$$

$$f^{(4)}(0) = 1$$

$$\therefore \cos x = 1 + x \cdot 0 + \frac{x^2}{2!} \cdot (-1) + \frac{x^3}{3!} \cdot 0 + \frac{x^4}{4!} \cdot 1 + \dots$$

$$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

(vi) $\tan^{-1} x$

$$f(x) = \tan^{-1} x$$

$$f(0) = 0$$

$$f'(x) = \frac{1}{1+x^2}$$

$$f'(0) = 1$$

$$f''(x) = \frac{(1+x^2) \cdot 0 - 1(2x)}{(1+x^2)^2} = \frac{-2x}{(1+x^2)^2}$$

$$f''(0) = 0$$

$$f'''(x) = \frac{-2(1+x^2)^2 - 2x(2)(1+x^2)(2x)}{(1+x^2)^4} = \frac{-2(1+x^2)^2 - 4x^2(1+x^2)}{(1+x^2)^4}$$

$$f'''(0) = -2$$

(vii) $e^{\sin x}$.

$$f(x) = e^{\sin x} \quad f'(0) = e^{\sin 0} = 1$$

$$f'(x) = e^{\sin x} \cdot \cos x \quad f'(0) = 1$$

$$f''(x) = f'(x) \cos x - \sin x f(x) \quad f''(0) = 1$$

$$f'''(x) = f''(x) \cos x - \cancel{\cos x}^{\sin x} f'(x) - \cos x f(x) \quad f'''(0) = 0$$

$$\therefore e^{\sin x} = 1 + x + \frac{x^2}{2!} + \dots$$

> Partial differentiation:

$$\boxed{Z = x^2y + x + y}$$

A variable z which has a definite value for every pair of values x and y is called a function of 2 independent variables x and y and we denote $z = f(x, y)$ or $\phi(x, y)$.

Let $z = f(x, y)$ be a function of 2 variable x and y .

The derivative of z w.r.t. x treating y as a constant is called the partial derivatives of z w.r.t. x and it is denoted by

$$\frac{\partial z}{\partial x}, \frac{\partial f}{\partial x}, f_x, \dots$$

The derivative of z w.r.t y keeping x as constant is called the partial derivative of z w.r.t y and it is denoted by

$$\frac{\partial z}{\partial y}, \frac{\partial f}{\partial y}, f_y, \dots$$

Note: $\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial^2 z}{\partial x^2}$

$$\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial^2 z}{\partial y^2}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial^2 z}{\partial y \partial x}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial^2 z}{\partial x \partial y}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$$

Ex: (i) Find 1 and 2 partial derivatives of $z = x^3 + y^3 - 3axy$

$$\rightarrow z = x^3 + y^3 - 3axy$$

$$\frac{\partial z}{\partial x} = 3x^2 + 0 - 3ay$$

$$\frac{\partial z}{\partial y} = 3y^2 - 3ax$$

$$\frac{\partial^2 z}{\partial x^2} = 6x, \quad \frac{\partial^2 z}{\partial y^2} = 6y$$

$$\frac{\delta^2 z}{\delta y \delta x} = -3a$$

$$\frac{\delta^2 z}{\delta y \delta x} = -3a.$$

(ii) If $u = \tan^{-1} \left(\frac{y}{x} \right)$ $\frac{\delta^2 u}{\delta x \delta y} = \frac{\delta^2 u}{\delta y \delta x}.$

$$\frac{\delta u}{\delta x} = \frac{1}{1 + \left(\frac{y}{x} \right)^2} \cdot \left(-\frac{y}{x^2} \right)$$

with respect to x so y is constant
then diff of $1/x$

$$\frac{\partial u}{\partial x} = \frac{-y}{x^2 + y^2}.$$

$$\frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial^2 u}{\partial y \partial x} = \frac{(x^2 + y^2)(-1) - (y)(2y)}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{y^2 - x^2}{(x^2 + y^2)^2} \rightarrow \textcircled{1}$$

$$\frac{\partial u}{\partial y} = \frac{1}{1 + \frac{y^2}{x^2}} \cdot \frac{1}{x} = \frac{x}{x^2 + y^2}$$

$$\frac{\partial^2}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial^2 u}{\partial x \partial y} = \frac{(x^2 + y^2)(1) - (x)(2x)}{(x^2 + y^2)^2}.$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{x^2 + y^2 - 2x^2}{(x^2 + y^2)^2} =$$

$$\frac{-x^2 + y^2}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2} \rightarrow \textcircled{2}$$

From $\textcircled{1}$ and $\textcircled{2}$
it is proved //

(iii) If $u = \phi(y+ax) + \psi(y-ax)$

$$\frac{\partial^2 u}{\partial x^2} = a^2 \frac{\partial^2 u}{\partial y^2}$$

$$\frac{\partial u}{\partial x} = \phi'(y+ax) \cdot a + \psi'(y-ax) (-a)$$

$$\frac{\partial u}{\partial x^2} = \phi''(y+ax)(a^2) + \psi''(y-ax)(a)^2 \rightarrow \textcircled{1}$$

$$\frac{\partial^2 u}{\partial y^2} = \phi''(y+ax) \cdot 1 + \psi''(y-ax) \cdot 1 \rightarrow \textcircled{2}$$

From $\frac{\partial^2 u}{\partial y^2}$

$$\textcircled{1} \rightarrow \frac{\partial^2 u}{\partial x^2} = a^2 [\phi''(y+ax) + \psi''(y-ax)]$$

$$\frac{\partial^2 u}{\partial x^2} = a^2 \frac{\partial^2 u}{\partial y^2} \text{ from } \textcircled{2}$$

(iv) If $z = e^{ax+by} f(ax-by)$

P.T $b \frac{\partial z}{\partial x} + a \frac{\partial z}{\partial y} = 2abz$

$$\frac{\partial z}{\partial x} = e^{ax+by} (a) f(ax-by) + e^{ax+by} f'(ax-by) a$$

$$\frac{\partial z}{\partial y} = e^{ax+by} (b) f(ax-by) + e^{ax+by} f'(ax-by) (-b)$$

$$a \frac{\partial z}{\partial x} = ab \{ e^{ax+by} f(ax-by) + e^{ax+by} f'(ax-by) \}$$

$$a \frac{\partial z}{\partial y} = ab \{ e^{ax+by} f(ax-by) - e^{ax+by} f'(ax-by) \}$$

$$b \frac{\partial^2 z}{\partial x^2} + a \frac{\partial^2 z}{\partial y^2} = 2ab e^{ax+by} f(ax-by)$$

$$= 2abz \quad \text{Hence proved...}$$

(v) If $u = \frac{e^{x+y}}{e^x + e^y}$ then s.t. $u_x + u_y = u$

$$\frac{\partial u}{\partial x} = \frac{e^{x+y} (e^x + e^y) - e^{x+y} e^x}{(e^x + e^y)^2}$$

$$u_x = \frac{e^{x+y} e^y}{(e^x + e^y)^2}$$

$$\rightarrow u_y = \frac{(e^x + e^y) e^{x+y} - e^{x+y} e^y}{(e^x + e^y)^2}$$

$$u_y = \frac{e^{x+y} e^x}{(e^x + e^y)^2}$$

$$u_x + u_y = \frac{e^{x+y} (e^x + e^y)}{(e^x + e^y)^2} = \frac{e^{x+y}}{e^x + e^y} //$$

Hence proved.

(vi) If $f = \log \left(\frac{x+y}{xy} \right)$ verify that $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$

$$\Rightarrow f = \log(x+y) - \log xy$$

$$f = \log(x+y) - [\log x + \log y]$$

$$= \log(x+y) - \log x - \log y$$

$$\frac{\partial f}{\partial x} = \frac{1}{x+y} - \frac{1}{x} = 0$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{(x+y)^0 - 1}{(x+y)^2} = 0 = \frac{-1}{(x+y)^2}$$

$$\frac{\partial f}{\partial y} = \frac{1}{x+y} - \frac{1}{y} = 0$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{-1}{(x+y)^2}$$

$$\therefore \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} \quad \text{hence proved}$$

• Total derivation:

If $u = f(x, y)$ where $x = \phi(t)$ and $y = \psi(t)$ then we can express 'u' as function of 't' alone by substituting the values of x and y in $f(x, y)$ thus we can find the ordinary derivative $\frac{du}{dt}$ which is called the total derivative of 'u' and is given by,

$$\boxed{\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}}$$

If $z = f(x, y)$ where $x = \phi(u, v)$ and $y = \psi(u, v)$ then we can express 'z' as a function of 'u' and 'v' or find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ i.e.

$$\checkmark \quad \frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\checkmark \quad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

(i) If $u = e^x \sin(yz)$

where $x = t^2$, $y = t-1$, $z = 1/t$

Find $\frac{du}{dt}$ at $t=1$

Sol: $u \rightarrow x, y, z \rightarrow t$

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt}$$

$$= e^x \sin(yz) \cdot 2t + e^x \cos(yz) \cdot 1 + e^x \cos(yz) \cdot y \cdot (-1/t^2)$$

$$\frac{du}{dt} = e^{t^2} \sin(t-1 \cdot 1/t) \cdot 2t + e^{t^2} \cos(t-1 \cdot 1/t) \cdot 1 + e^{t^2} \cos(t-1 \cdot 1/t) \cdot (t-1) \cdot (-1/t^2)$$

$t=1$

$$= e \sin(0) \cdot 2 + e^1 \cos(0) \cdot 1 + e^1 \cos(0) \cdot (-1/1^2)$$

$$= 0 + e + 0 = e //$$

(ii) If $u = x^2 - y^2$

where $x = 2r - 3s + 4$, $y = -r + 8s - 5$

Find $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial s}$

Sol: $u \rightarrow x, y \rightarrow r, s$

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r} \rightarrow \textcircled{1}$$

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial s} \rightarrow \textcircled{2}$$

$$\frac{\partial u}{\partial x} = 2x \cdot 2 - 2y \quad (1)$$

$$4x + 2y = 4[2x - 3z + 4] + 2[-x + 8z - 5]$$

$$\frac{\partial u}{\partial x} = 6x + 4z + 6 //$$

$$\frac{\partial u}{\partial z} = 2x \cdot (-3) - 2y \cdot 8$$

$$= -6x - 16y$$

$$= -6(2x - 3z + 4) - 16(-x + 8z - 5)$$

$$= -12x + 18z + 24 + 16x - 128z - 80 + 80$$

$$= \underline{\underline{4x - 110z + 56}}$$

HW
(*) $f = \log \left(\frac{x+y}{xy} \right)$

$$\rightarrow f = (\log(x+y) - [\log x + \log y])$$

$$= \log(x+y) - \log x - \log y$$

$$\frac{\partial f}{\partial x} = \frac{1}{x+y} - \frac{1}{x} = 0$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{(x+y)^0 \cdot 1}{(x+y)^2} - 0 = \frac{-1}{(x+y)^2}$$

$$\frac{\partial f}{\partial y} = \frac{1}{x+y} - \frac{1}{y} = 0$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{-1}{(x+y)^2}$$

$$\therefore \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} \quad \text{hence proved}$$

(ii) If $z = f(u, v)$ where $u = x^2 + y^2$, $v = 2xy$.
Prove that

$$(i) \quad x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = 2(x^2 + y^2) \frac{\partial z}{\partial u}$$

$$(ii) \quad \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 = 4(x^2 + y^2) \left[\left(\frac{\partial z}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2 \right]$$

$$\rightarrow z \rightarrow u, v \rightarrow x, y$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot 2x + \frac{\partial z}{\partial v} \cdot 2y \rightarrow (1)$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot (2y) + \frac{\partial z}{\partial v} \cdot 2x \rightarrow (2)$$

$$(1) \times x - (2) \times y$$

$$x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = 2x^2 \frac{\partial z}{\partial u} + 2xy \frac{\partial z}{\partial v} - 2xy \frac{\partial z}{\partial v} - 2y^2 \frac{\partial z}{\partial u}$$

$$= 2(x^2 + y^2) \frac{\partial z}{\partial u} //$$

$$\begin{aligned} \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 &= \left(\frac{\partial z}{\partial u} \right)^2 (4x^2) + \left(\frac{\partial z}{\partial v} \right)^2 (4y^2) + 2 \left(\frac{\partial z}{\partial u} \right)^2 (4xy^2) + 2 \left(\frac{\partial z}{\partial v} \right)^2 (4x^2y) \\ &= \left(\frac{\partial z}{\partial u} \right)^2 (4x^2 + 4y^2 + 8xy^2/x + 8x^2y/y) \\ &= \left(\frac{\partial z}{\partial u} \right)^2 (4x^2 + 4y^2 + 8xy) \end{aligned}$$

$$= \left(\frac{\partial z}{\partial u} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial v} \frac{\partial y}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial u} \frac{\partial y}{\partial v} + \frac{\partial z}{\partial v} \frac{\partial x}{\partial u} \right)^2$$

$$= \left(\frac{\partial z}{\partial u} \right)^2 4x^2 + \left(\frac{\partial z}{\partial v} \right)^2 4y^2 +$$

$$8 \left(\frac{\partial z}{\partial u} \right) \left(\frac{\partial z}{\partial v} \right) + \left(\frac{\partial z}{\partial u} \right)^2 (4y)^2 + \left(\frac{\partial z}{\partial v} \right)^2 (4x)^2$$

$$= \left[\left(\frac{\partial z}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2 \right] 4(x^2 + y^2)$$

(iii) If $u = f\left(x, y, \frac{y}{x}\right)$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{y}{x} \frac{\partial u}{\partial \left(\frac{y}{x}\right)}$

Sol:

$$u = f(r, s)$$

$$r = x, \quad s = \frac{y}{x}$$

$$u \longrightarrow r, s \longrightarrow x, y, z.$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x}$$

$$= \frac{\partial u}{\partial r} \cdot x + \frac{\partial u}{\partial s} (0) \longrightarrow \textcircled{1}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial y}$$

$$= \frac{\partial u}{\partial r} \cdot 0 + \frac{\partial u}{\partial s} \cdot \frac{1}{x} \longrightarrow \textcircled{2}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial z}$$

$$= \frac{\partial u}{\partial r} x + \frac{\partial u}{\partial s} \frac{-y}{x^2} \longrightarrow \textcircled{3}$$

$$\rightarrow x \left(\frac{\partial u}{\partial r} \right) x - y \left(\frac{\partial u}{\partial s} \frac{1}{z} \right) - z \left(\frac{\partial u}{\partial r} x + \frac{\partial u}{\partial s} \left(-\frac{y}{z^2} \right) \right)$$

$$\rightarrow -y \left(\frac{\partial u}{\partial s} \frac{1}{z} \right) + z \frac{\partial u}{\partial s} \frac{y}{z^2}$$

$$\rightarrow -\frac{y}{z} \frac{\partial u}{\partial s} + \frac{\partial u}{\partial s} \frac{y}{z} = 0 \quad \text{Hence proved.}$$

(iv) If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$

Show that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.

Sol: $u = f(r, s)$

$$r = \frac{y-x}{xy} \rightarrow s = \frac{z-x}{xz} = \frac{1}{x} \frac{1}{z} - \frac{1}{z}$$

$$u \rightarrow r, s \rightarrow x, y, z$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x}$$

$$= \frac{\partial u}{\partial r} \cdot \left(-\frac{1}{x^2} \right) + \frac{\partial u}{\partial s} \left(-\frac{1}{x^2} \right) \rightarrow (1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial y}$$

$$= \frac{\partial u}{\partial r} \left(\frac{1}{y^2} \right) + \frac{\partial u}{\partial s} \left(\frac{1}{y^2} \right) \rightarrow (2)$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial z}$$

$$= \frac{\partial u}{\partial r} (0) + \frac{\partial u}{\partial s} \left(-\frac{1}{z^2} \right) \rightarrow (3)$$

$$\rightarrow x^2 \left(-\frac{\partial u}{\partial r} \cdot \frac{1}{x^2} - \frac{\partial u}{\partial s} \frac{1}{x^2} \right) + y^2 \left(\frac{\partial u}{\partial r} \left(\frac{1}{y^2} \right) + \frac{\partial u}{\partial s} \left(\frac{1}{y^2} \right) \right) + z^2 \left(\frac{\partial u}{\partial r} (0) - \frac{\partial u}{\partial s} \frac{1}{z^2} \right) = 0 \quad \text{Hence proved}$$

~~thw~~
~~(ix) if $x = f(x, y)$~~

> Differentiation of Implicit function:

If $f(x, y) = a$ be an implicit relation b/w x and y , which defines as differential function of x ,

$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$$

ii) If $xy^3 - yx^3 = 6$ is the equation of curve. Find the slope and equation of tangent line at the point $(1, 2)$.

Sol: slope $\left[\frac{dy}{dx} = \frac{-\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \right]$

$$\frac{dy}{dx} = \frac{-[y^3 - 3x^2y]}{[3y^2x - x^3]}$$

$$\text{at } (1, 2) \frac{dy}{dx} = -\frac{2}{11}$$

Equation of the tangent line

$$y - y_1 = m(x - x_1)$$

$$y - 2 = \frac{-2}{11}(x - 1)$$

7 Jacobians:

If u, v are functions of 2 independent variables x, y then,

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

is called Jacobian of (u, v) w.r.t (x, y) .

Similarly, The Jacobian of u, v, w w.r.t (x, y, z)

$$J = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

Properties of Jacobians:

- (i) If $J = \frac{\partial(u, v)}{\partial(x, y)}$ and $J' = \frac{\partial(x, y)}{\partial(u, v)}$ then $\boxed{J \times J' = 1}$
- (ii) If (u, v) are functions of (r, s) and (r, s) are functions of (x, y) then,

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \frac{\partial(u, v)}{\partial(r, s)} \cdot \frac{\partial(r, s)}{\partial(x, y)} //$$

- 1) If $x = r \cos \theta$ $y = r \sin \theta$. Find $J = \frac{\partial(x, y)}{\partial(r, \theta)}$ and

$$J' = \frac{\partial(r, \theta)}{\partial(x, y)}$$

$$\text{and ST } JJ' = 1$$

sol: $J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$

$$= \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$= r \cos^2 \theta + r \sin^2 \theta$$

$$= r //$$

then $x^2 + y^2 = r^2$

$$r = \sqrt{x^2 + y^2}$$

$$\frac{y}{x} = \tan \theta$$

$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$

we need (r, θ) to be functions of x, y
Hence.

$$J' = \frac{\partial(r, \theta)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{1}{\sqrt{x^2 + y^2}} \cdot x & \frac{1}{\sqrt{x^2 + y^2}} \cdot y \\ \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \left(-\frac{y}{x^2}\right) & \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{1}{x} \end{vmatrix}$$

$$= \frac{x}{\sqrt{x^2 + y^2}} \left(\frac{y}{(x^2 + y^2)x} \right) + \frac{y}{\sqrt{x^2 + y^2}} \left(\frac{y}{x^2 + y^2} \right)$$

$$= \frac{x^2}{\sqrt{x^2 + y^2} (x^2 + y^2)} + \frac{y^2}{\sqrt{x^2 + y^2} (x^2 + y^2)}$$

$$\rightarrow \frac{x^2 + y^2}{(x^2 + y^2) \left[\sqrt{x^2 + y^2} \right]}$$

$$\rightarrow \frac{1}{\sqrt{x^2 + y^2}} = \frac{1}{r} = J' //$$

$$\therefore JJ' = r \times \frac{1}{r} = 1 //$$

2. If $u = x^2 - 2y^2$, $v = 2x^2 - y^2$, where $x = r \cos \theta$, $y = r \sin \theta$

show that $\frac{\partial(u,v)}{\partial(r,\theta)} = 6r^3 \sin 2\theta$

Sol:

$$u, v \rightarrow x, y \rightarrow r, \theta$$

$$\frac{\partial(u,v)}{\partial(r,\theta)} = \frac{\partial(u,v)}{\partial(x,y)} \frac{\partial(x,y)}{\partial(r,\theta)}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} 2x & -4y \\ 4x & -2y \end{vmatrix} \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$= (-4xy + 16xy) (r \cos^2 \theta \sin \theta + r (\sin^2 \theta \cos \theta))$$

$$= 12xy \sin 2\theta \cdot r (\cos^2 \theta \sin \theta + \sin^2 \theta \cos \theta)$$

$$= 12(r \cos \theta)(r \sin \theta) \sin 2\theta \cdot r$$

$$= 12r^3 \sin 2\theta$$

$$= 6r^3 \sin 2\theta //$$

$$3. \text{ If } u = x + \frac{y^2}{x} \quad v = \frac{y}{x}.$$

$$p.1 \quad JJ' = 1.$$

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} 1 - \frac{y^2}{x^2} & \frac{2y}{x} \\ -\frac{y^2}{x^2} & \frac{2y}{x} \end{vmatrix}$$

$$= \left(1 - \frac{y^2}{x^2}\right) \left(\frac{2y}{x}\right) - \frac{2y}{x} \left(-\frac{y^2}{x^2}\right)$$

$$= \frac{2y}{x} - \frac{2y^3}{x^3} + \frac{2y^3}{x^3}$$

$$\rightarrow \frac{x^2 - y^2 + y^2}{x^3} = \frac{x^2}{x^3} = \frac{2y}{x} //$$

$$\rightarrow J' = \frac{\partial(x,y)}{\partial(u,v)}$$

$$u - v = \left(x + \frac{y^2}{x} - \frac{y^2}{x}\right) = x.$$

$$\boxed{u - v = x}$$

$$v = \frac{y^2}{u - v}$$

$$\boxed{y = \sqrt{v(u-v)}}$$

$$J' = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} +1 & -1 \\ \frac{1}{2\sqrt{v(u-v)}} \cdot v & \frac{1}{2\sqrt{v(u-v)}} \cdot (u-2v) \end{vmatrix}$$

$$= + \frac{v}{2\sqrt{v(u-v)}} + \frac{(u-2v)}{2\sqrt{v(u-v)}}$$

$$= \frac{x}{2y} //$$

$$JJ' = 1 //$$

4. If $x = e^u \sec v$, $y = e^u \tan v$ P.I $JJ' = 1$

$\rightarrow x, y \rightarrow u, v$

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$J = \begin{vmatrix} e^u \sec v & e^u \sec v \tan v \\ e^u \tan v & e^u \sec^2 v \end{vmatrix}$$

$$J = e^{2u} \sec^3 v - e^{2u} \tan^2 v \sec v$$

$$J = e^{2u} \sec v (\sec^2 v - \tan^2 v)$$

$$J = e^{2u} \sec v (1) //$$

$$J' = \frac{\partial(u,v)}{\partial(x,y)}$$

So here $u, v \rightarrow x, y$
for that in terms of u , $x^2 - y^2 = e^{2u} \sec^2 v - e^{2u} \tan^2 v$
 $= e^{2u}$

And here exponential
we are taking log.

$$\rightarrow \log(x^2 - y^2) = 2u.$$

$$\therefore \boxed{u = \frac{1}{2} \log(x^2 - y^2)}$$

Now in terms of v ,

$$\frac{y}{x} = \frac{e^u \tan v}{e^u \sec v} = \sin v.$$

$$\therefore \boxed{v = \sin^{-1}\left(\frac{y}{x}\right)}$$

$$J' = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} \frac{1}{x^2 - y^2} \cdot 2x & \frac{1}{2} \frac{1}{x^2 - y^2} \cdot (-2y) \\ \frac{1}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} \cdot \left(-\frac{y}{x^2}\right) & \frac{1}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} \cdot \frac{1}{x} \end{vmatrix}$$

$$= \frac{1}{2} \cdot \frac{1}{x^2 - y^2} \cdot \frac{x}{\sqrt{x^2 - y^2}} - \frac{y^2}{\sqrt{x^2 - y^2} (x^2 - y^2)} \cdot \frac{1}{x}$$

$$= \frac{x^2 - y^2}{(x^2 - y^2)^2 (\sqrt{x^2 - y^2})} = \frac{1}{e^{2u} \sec^2 v \cdot e^u e^{2u} \sec^2 v} = \frac{1}{e^{5u} \sec^4 v}$$

> Taylor's series of 2 variables:

$$f(x, y) = f(a, b) + \left[(x-a) f_x(a, b) + (y-b) f_y(a, b) \right] + \frac{1}{2!} \left[(x-a)^2 f_{xx}(a, b) + 2(x-a)(y-b) f_{xy}(a, b) + (y-b)^2 f_{yy}(a, b) \right] + \frac{1}{3!} \left[(x-a)^3 f_{xxx}(a, b) + 3(x-a)^2(y-b) f_{xxy}(a, b) + 3(x-a)(y-b)^2 f_{xyy}(a, b) + (y-b)^3 f_{yyy}(a, b) \right] + \dots$$

> Maclaurin's series:

$a=0, b=0$ in Taylor's series.

$$f(x, y) = f(0) + \left[(x) f_x(0) + (y) f_y(0) \right] + \frac{1}{2!} \left[(x)^2 f_{xx}(0) + 2(x)(y) f_{xy}(0) + (y)^2 f_{yy}(0) \right] + \frac{1}{3!} \left[(x)^3 f_{xxx}(0) + 3(x)^2(y) f_{xxy}(0) + 3(x)(y)^2 f_{xyy}(0) + (y)^3 f_{yyy}(0) \right] + \dots$$

(i) Find the Taylor series of the function $f(x, y) = x^2 + xy + y^2$ at $(1, 2)$.

$$f(x, y) = f(1) + f'(1, 2)$$

$$f(x, y) = x^2 + xy + y^2$$

$$f_{xx} = 2$$

$$f_{xxx} = 0$$

$$f_y = x + 2y$$

$$f_{yy} = 2$$

$$f_{yyy} = 0$$

$$f_{xy} = 1$$

$$f_{xxy} = 0$$

$$f_{xyy} = 0$$

$$f_x(1,2) = 4.$$

$$f_{xx}(1,2) = 2$$

$$f_{xxx}(1,2) = 0$$

$$f_y(1,2) = 5$$

$$f_{yy}(1,2) = 2$$

$$f_{yyy}(1,2) = 0$$

$$f(x,y) = x^2 + xy + y^2 = 7 + [(x-1)4 + (y-2)5] + \frac{1}{2!} [(x-1)^2 2 + 2(x-1)(y-2) + (y-2)^2 2] +$$

$$\frac{1}{3!} [(x-1)^3 0 + 3(x-1)^2(y-2) 0 + 3(x-1)(y-2)^2 0 + (y-2)^3 0]$$

$$x^2 + xy + y^2 = 7 + [4x - 4 + 5y - 10] + \frac{1}{2!}$$

$$[2x^2 + 2 \cdot 4x + 2x - 2(y-2) + 2y^2 + 2 \cdot 4y]$$

$$\frac{1}{3!} [0] //$$

(ii) Expand the function $f(x, y) = \sin(xy)$ at the point $(1, \pi/2)$ upto second degree terms

$$f_x(x, y) = \cos(xy) \cdot y$$

$$f_y(x, y) = \cos(xy) \cdot x$$

$$f_{xx}(x, y) = -\sin(xy) \cdot y^2$$

$$f_{yy}(x, y) = -\sin(xy) \cdot x^2$$

$$f_{xy}(x, y) = \cos(xy) \cdot y - \cos(xy) \cdot x$$

at the point $(1, \pi/2)$ upto

$$f(1, \pi/2) = 1$$

$$f_x(1, \pi/2) = 0$$

$$f_y(1, \pi/2) = 0$$

$$f_{xx}(1, \pi/2) = -\pi^2/4$$

$$f_{yy}(1, \pi/2) = -1$$

$$f_{xy}(1, \pi/2) = -1$$

$$\begin{aligned} \sin(xy) &= f(1, \pi/2) + \left[(x-1) \cdot 0 + (y-\pi/2) \cdot 0 \right] + \frac{1}{2!} \\ &\quad \left[(x-1)^2 \left(-\frac{\pi^2}{4} \right) + 2(x-1)(y-\pi/2) \cdot (-1) + (y-\pi/2)^2 \cdot (-1) \right] \\ &= f(1, \pi/2) + \frac{1}{2!} \left[-\frac{(x-1)^2 \pi^2}{4} + 2(x-1)(y-\pi/2) - (y-\pi/2)^2 \right] \end{aligned}$$

i) Expand e^{xy} at $(1, 1)$ in Taylor series upto terms of second degree.

$$f_x(x,y) = e^{xy} \cdot (y)$$

$$f_y(x,y) = e^{xy} \cdot (x)$$

$$f_{xx}(x,y) = e^{xy} \cdot y^2$$

$$f_{yy}(x,y) = e^{xy} \cdot x^2$$

$$f_{xy}(x,y) = e^{xy} (y) - e^{xy} \cdot (x)$$

$$f_x(1,1) = e$$

$$f_y(1,1) = e$$

$$f_{xx}(1,1) = e$$

$$f_{yy}(1,1) = e$$

$$f_{xy} = 0$$

$$f(2) + [(x-1)e + (y-1)e]$$

(iv) $e^{xy} = 0 + \left[\cancel{xy} \right] + \frac{1}{2!} [(x-1)^2 e + 2(x-1)(y-1)e + (y-1)^2 e] + \frac{1}{3!} [(x-1)^3 + \dots]$

(iv) expand the following terms upto third degree terms:

(a) $f(x,y) = e^y \log(1+x)$

$$f_x(x,y) = e^y \frac{1}{1+x} \cdot (1)$$

$$f_x(0,0) = 1$$

$$f_{xy}(x,y) = e^y \log(1+x)$$

$$f_y(0,0) = 0$$

$$f_{xx}(x,y) = e^y \left(\frac{-1}{(1+x)^2} \right)$$

$$f_{xx}(0,0) = -1$$

$$f_{yy}(x,y) = e^y \log(1+x)$$

$$f_{yy}(0,0) = 0$$

$$f_{xy}(x,y) = e^y \frac{1}{1+x}$$

$$f_{xy}(0,0) = 1$$

$$f_{xxx}(x,y) = e^y \left(\frac{2}{(1+x)^3} \right) \quad f_{xxx}(0,0) = 2$$

$$f_{xy}(x,y) = e^y \left(\frac{-1}{(1+x)^2} \right) \quad f_{xy}(0,0) = -1$$

$$f_{yy}(x,y) = e^y \frac{1}{1+x} \quad f_{yy}(0,0) = 1$$

$$f_{yyy}(x,y) = \log(1+x) e^y \quad f_{yyy}(0,0) = 0$$

$$f(x,0) = f(0,0) + f'(x,0)$$

$$e^y \log(1+x) = 0 + \left[x f'_x(0,0) + y(0) \right] + \frac{1}{2!} \left[x^2 f''_{xx}(0,0) + 2xy f''_{xy}(0,0) + y^2 f''_{yy}(0,0) \right] + \frac{1}{3!} \left[x^3 f'''_{xxx}(0,0) + 3xy^2 f'''_{xyy}(0,0) + y^3 f'''_{yyy}(0,0) \right]$$

$$e^y \log(1+x) = x - \frac{x^2}{2} + xy + \frac{x^3}{3} - \frac{x^2 y}{2} + \frac{xy^2}{2} + \dots //$$

(b) $\cos x \cos y$

$$f_x(x,y) = -\sin x \cos y$$

$$f_y(x,y) = \cos x - \sin y$$

$$f_{xx}(x,y) = -\cos x \cos y$$

$$f_{yy}(x,y) = \cos x - \cos y$$

$$f_{xy}(x,y) = -\sin x \cos y - \cos x \sin y$$

$$f_x(0,0) = 0$$

$$f_y(0,0) = 0$$

$$f_{xx}(0,0) = -1$$

$$f_{yy}(0,0) = -1$$

$$f_{xy}(0,0) = 0$$

$$f_{xx}(x,y) = \sin x \cos y$$

$$f_{xx}(a,0) = 0.$$

$$f_{yy}(x,y) = \cos x \sin y$$

$$f_{yy}(0,0) = 0$$

$$f_{xy}(x,y) = -\cos x - \sin y$$

$$f_{xy}(0,0) = 0$$

$$f_{yx}(x,y) = -\sin x - \cos y$$

$$f_{yx}(x,y) = 0$$

$$\begin{aligned} \cos x \cos y &= 0 + [(x) + 0] + \frac{1}{2!} [(x)^2 (-1) + \cancel{2xy(x)}] + \\ &\quad \frac{1}{3!} [x^3 + 2(x)(y)0 + (y)^2(-1)] + \\ &\quad \frac{1}{3!} [(x)^3 0 + 3(x)^2 y (0) + 3xy^2 (0) + y^3 0] \\ &= x + \frac{1}{2!} [-x^2 - y^2] + \dots // \end{aligned}$$

$$(d) \quad e^{ax+by}$$

$$f_x(x, y) = e^{ax+by} \cdot (a)$$

$$f_y(x, y) = e^{ax+by} \cdot (b)$$

Maximum and Minimum for a function of two variables:

A function $f(x, y)$ is said to have maximum value at $x = a, y = b$, if $f(a, b)$ is greater than $f(a+h, b+k)$ for small values of h and k (h and k are +ve or -ve).

A function $f(x, y)$ is said to have minimum value at $x = a, y = b$, if $f(a, b)$ is less than $f(a+h, b+k)$.

A maximum or minimum value of a function is called an extreme value.

Saddle point or Minimax:

It is a point where function is neither maximum nor minimum. It is a such a point max in one direction while ~~max~~ minimum in another direction.

The necessary condition for $f(x, y)$ to have maximum or minimum at (a, b) are,

$$f_x(a, b) = 0$$

$$f_y(a, b) = 0$$

Working rule to find the maximum and minimum values:

$$Z = f(x, y)$$

- * Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
- * Solve $\frac{\partial z}{\partial x} = 0$ and $\frac{\partial z}{\partial y} = 0$ simultaneously.
- * Let (a, b) , (c, d) be the solutions of these equations.
- * For each solution in step 2. Find $r = \frac{\partial^2 z}{\partial x^2}$, $s = \frac{\partial^2 z}{\partial x \partial y}$, $t = \frac{\partial^2 z}{\partial y^2}$.
- * (a) If $rt - s^2 < 0$ for particular solution (a, b) of step 2 then z has no extreme value at (a, b) (Saddle point)
- (b) If $rt - s^2 > 0$ and $r < 0$ for a particular solution (a, b) of step 2 then z has maximum value at (a, b)
- (c) If $rt - s^2 > 0$ and $r > 0$ for a particular solution (a, b) of step 2 then z has a minimum value at (a, b)

Note: (1) If $rt - s^2 = 0$ at (a, b) the case is doubtful and needs further investigation.

(2) The points (a, b) , (c, d) are called stationary points and the values of $f(x, y)$ at these points are called stationary values.

- 1) Find extreme values of $x^2 + y^2 + 6x - 12$.

$$z = x^2 + y^2 + 6x - 12$$

$$\frac{\partial z}{\partial x} = 2x + 0 + 6 = 2x + 6$$

$$\frac{\partial z}{\partial y} = 2y$$

$$r = \frac{\partial^2 z}{\partial x^2} = 2$$

$$s = \frac{\partial^2 z}{\partial x \partial y} = 0$$

$$t = \frac{\partial^2 z}{\partial y^2} = 2$$

$$\frac{\partial z}{\partial x} = 0 \text{ and } \frac{\partial z}{\partial y} = 0.$$

$$2x + 6 = 0$$

$$2y = 0.$$

$$\boxed{y = 0}$$

$$2x = -6$$

$$\boxed{x = -3}$$

$\therefore (-3, 0)$ is a stationary point

$$rt - s^2 = 2(2) - 0 = 4 > 0.$$

$$r = 2 > 0.$$

$\therefore$ Hence z has a minimum value at $(-3, 0)$

$$z = (-3)^2 + 0 + 6(-3) - 12 \\ = -21 //$$

2) Find the extreme values of following:

i) $f = 1 - x^2 - y^2$

$$\frac{\partial f}{\partial x} = -2x$$

$$r = \frac{\partial^2 f}{\partial x^2} = -2$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$\frac{\partial f}{\partial y} = -2y$$

$$t = \frac{\partial^2 f}{\partial y^2} = -2$$

$$-2x = 0$$

$$\boxed{x = 0}$$

and

$$-2y = 0$$

$$\boxed{y = 0}$$

$\therefore (0,0)$ is stationary point.

$$rt^2 - s^2 = -2(-2)^2 - 0 = +4 > 0$$

$$r = -2 < 0$$

$\therefore f$ has ~~no~~ maximum extreme value at $(0,0)$

$$z: 1 - (2)^2 - (2)^2$$

$$= 1 - 4 - 4$$

$$= -7$$

(ii) $f: x^2 + y^2$

$$\frac{\partial f}{\partial x} = 2x \quad \frac{\partial f}{\partial y} = 2y$$

$$r = \frac{\partial^2 f}{\partial x^2} = 2$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$t = \frac{\partial^2 f}{\partial y^2} = 2$$

$$\frac{\partial f}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0$$

$$2x = 0$$

$$\boxed{x=0}$$

$$2y = 0$$

$$\boxed{y=0}$$

$\therefore (0,0)$ is stationary point

$$rt - s^2 = 2(2) - (0)^2 = +4 > 0$$

$$= 2 > 0$$

$\therefore z$ has ~~no extreme~~ minimum value.

(iii) xy .

$$\frac{\partial f}{\partial x} = y \quad \frac{\partial f}{\partial y} = x$$

$$r = \frac{\partial^2 f}{\partial x^2} = 0$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = 1$$

$$t = \frac{\partial^2 f}{\partial y^2} = 0$$

$$\frac{\partial f}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0$$

$$\boxed{y=0} \quad \boxed{x=0}$$

$(0,0)$ is stationary point.

$$rt - s^2 = y(x) - 1$$

$$= xy - 1$$

$$= -1 < 0$$

$$= 0 < 0$$

$\therefore Z$ has no extreme value.

Find the shortest distance from origin to the surface

$$\frac{1}{\sqrt{2}} xyz^2 = 2$$

Let d be the distance from origin $(0,0,0)$ to any point (x,y,z) of the given surface then $d = \sqrt{x^2 + y^2 + z^2}$

Squaring:

$$d^2 = x^2 + y^2 + z^2$$

$$= x^2 + y^2 + \frac{2}{xy}$$

(we need the function in 2 variable hence substituting it).

$$f(x,y) = x^2 + y^2 + \frac{2}{xy}$$

$$f_x = 2x - \frac{2}{x^2 y}$$

$$f_y = 2y - \frac{2}{y^2 x}$$

$$r = f_{xx} = 2 + \frac{2}{x^3 y}$$

$$s = f_{xy} = \frac{2}{x^2 y^2}$$

$$t = f_{yy} = 2 + \frac{2}{y^3 x}$$

$$f_x = 0 \text{ and } f_y = 0$$

$$2x - \frac{2}{y^2} = 0 \text{ and } 2y - \frac{2}{x^2} = 0$$

$$2xyx^2 - 2 = 0$$

$$2(xy^2 - 1) = 0$$

$$x^3y - 1 = 0$$

$$x^3y = 1 \rightarrow (1)$$

$$x^3y - 1 = 0$$

$$\cancel{x^3y} - \cancel{x^3y} \quad (1) - (2)$$

$$x^3y - 1 - (xy^3 - 1) = 0$$

$$x^3y - 1 - xy^3 + 1 = 0$$

$$x^3y - xy^3 = 0$$

$$xy(x^2 - y^2) = 0$$

$$x^2 - y^2 = 0$$

$$x^2 = y^2$$

$$\therefore \boxed{x = \pm y}$$

Substitute in (1)

$$\boxed{x = y}$$

$$y^3y - 1 = 0$$

$$y^4 = 1$$

$$\boxed{y = \pm 1}$$

$$\boxed{x = \pm 1}$$

$\therefore (1, 1) (-1, -1)$ are stationary points.

$$rt - s^2$$

at (1,1)

$$r = 6$$

$$s = 2$$

$$t = 6$$

$$rt - s^2 = 6(6) - 4 = 32 > 0$$

$$6 > 0$$

$\therefore f$ has minimum value at (1,1)

At (-1,-1), $r=6$, $s=2$, $t=6$.

$$rt - s^2 = 32 > 0$$

$$6 > 0$$

$\therefore f$ has minimum value at (-1,-1)

$$z^2 = 2 \quad \text{as } xy \text{ value is } 1.$$

$$\sqrt{x^2 + y^2 + z^2} = \sqrt{1+1+2} = \sqrt{4}$$

$$\therefore d = \sqrt{4} = 2 //$$

Lagrange's Method of undetermined multipliers:

This method is used to find the stationary values of a function $f(x, y, z)$ when (x, y, z) are not all independent but are connected by some relation say $\phi(x, y, z) = 0$.

Working rule.

$$1) \text{ Write } F = f(x, y, z) + \lambda \phi(x, y, z)$$

Obtain the equations $\frac{\partial F}{\partial x} = 0$, $\frac{\partial F}{\partial y} = 0$, $\frac{\partial F}{\partial z} = 0$. Solve the above equations together with $\phi(x, y, z) = 0$. The values of x, y, z so obtaining will give the stationary value of $f(x, y, z)$.

(i) Find the stationary value of $x^2 + y^2 + z^2$ when $x + y + z = 3$ by Lagrange's multipliers.

$$F = x^2 + y^2 + z^2 + \lambda [x + y + z - 3a]$$

$$\frac{\partial F}{\partial x} = 2x + \lambda$$

$$\frac{\partial F}{\partial y} = 2y + \lambda$$

$$\frac{\partial F}{\partial z} = 2z + \lambda$$

$$2x + \lambda = 0 \text{ --- (1)} \quad 2y + \lambda = 0 \text{ --- (2)} \quad 2z + \lambda = 0 \text{ --- (3)}$$

$$2x = -\lambda$$

$$x = -\frac{\lambda}{2}$$

$$y = -\frac{\lambda}{2}$$

$$z = -\frac{\lambda}{2}$$

$$(1) - (2) : 2x = 2y$$

$$x = y$$

$$(2) - (3) : y = z$$

$$\therefore x = y = z$$

$$(1) \quad 3x = 3a$$

$$x = a = y = z$$

$$\rightarrow x^2 + y^2 + z^2 = a^2 + a^2 + a^2 = 3a^2 \quad \left\{ f(x, y, z) \right.$$

(iii) Write a rectangular box open at the top is to have volume of 32 cubic meter. Find the dimensions of the box requiring least material for its construction.

Sol: Let x, y, z be the edges of the box at 'S' be its surface then $S = xy + 2yz + 2zx \rightarrow (1)$
 as it is open if it was not then $2xy$.

$$\boxed{xyz = 32} \rightarrow (2)$$

$$F = xy + 2yz + 2zx + \lambda [xyz - 32]$$

$$\frac{\partial F}{\partial x} = y + 2z + \lambda [yz]$$

$$\frac{\partial F}{\partial y} = x + 2z + \lambda [xz]$$

$$\frac{\partial F}{\partial z} = 2y + 2x + \lambda [xy]$$

$$y + 2z + \lambda yz = 0$$

$$y - 2z - \lambda yz = 0 \rightarrow (3)$$

$$x + 2z + \lambda xz = 0$$

$$x - 2z - \lambda xz = 0 \rightarrow (4)$$

$$2y + 2x + \lambda xy = 0 \rightarrow (5)$$

$$\textcircled{3} \times x - \textcircled{4} \times y \Rightarrow xy + 2xz + \lambda yz - xy - 2xy - \lambda yz = 0$$

$$2xz - 2xy = 0$$

$$2xz = 2xy$$

$$\boxed{x=y} \quad 2z \neq 0.$$

$$\textcircled{4} \times x - \textcircled{5} \times z$$

$$xy + 2xz + \lambda yz - 2yz - 2xz - \lambda yz = 0$$

$$xy - 2xz = 0$$

$$xy = 2xz$$

$$y = 2z$$

$$\boxed{z = y/2}$$

From $\textcircled{2}$

$$xyz = 32$$

$$y \cdot \frac{y}{2} = 32$$

$$y^2 = 64$$

$$y = 4 = x \text{ and } z = 2$$

$\therefore$ Surface area will be minimum when $x = y = 4$ and $z = 2$

(iii) If perimeter of Δl_e is constant Δl_e is maximum when the Δl_e is equilateral. Prove that area of this

1. Let a, b, c be the sides of Δ .

$$2s = a + b + c \quad (\text{perimeter } 2s \text{ is constant})$$

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$A^2 = s(s-a)(s-b)(s-c) = f$$

$$\frac{\partial f}{\partial a} = s(s-b)(s-c) \quad \frac{\partial A}{\partial a} = \frac{1}{2} \frac{\partial A^2}{\partial a} = \frac{1}{2} s(s-b)(s-c)$$

$$F = s(s-a)(s-b)(s-c) + \lambda[a+b+c-2s]$$

$$\frac{\partial F}{\partial a} = s(s-b)(s-c)(-1) + \lambda[1] = 0 \quad \text{--- (1)}$$

$$\frac{\partial F}{\partial b} = s(s-a)(s-c)(-1) + \lambda[1] = 0 \quad \text{--- (2)}$$

$$\frac{\partial F}{\partial c} = s(s-a)(s-b)(-1) + \lambda[1] = 0 \quad \text{--- (3)}$$

$$(2) - (3)$$

$$s(s-b)(s-c)(-1) + s(s-a)(s-c)(-1) = 0$$

$$s(s-a)(-1) [(s-c) - (s-b)] = 0$$

$$s(s-c) [-s + b + s - a] = 0$$

$$b - a = 0$$

$$\boxed{b = a} //$$

$$(3) - (1)$$

$$s(s-a)(s-c)(-1) + s(s-a)(s-b) = 0$$

$$s(s-a) [-s + c + s - b] = 0$$

$$\therefore b=c$$

$$\therefore a=b=c$$

$$\therefore \textcircled{1} \Rightarrow 2s = a+b+c$$

$$2s = 3a$$

$$s = \frac{3a}{2}$$

$$a = \frac{2s}{3} = b = c$$

$$A = \sqrt{s \left(s - \frac{2s}{3} \right)^3}$$

$$\therefore A = \sqrt{s \left(s - \frac{2s}{3} \right)^{3/2}}$$

is maximum when $a=b=c$.